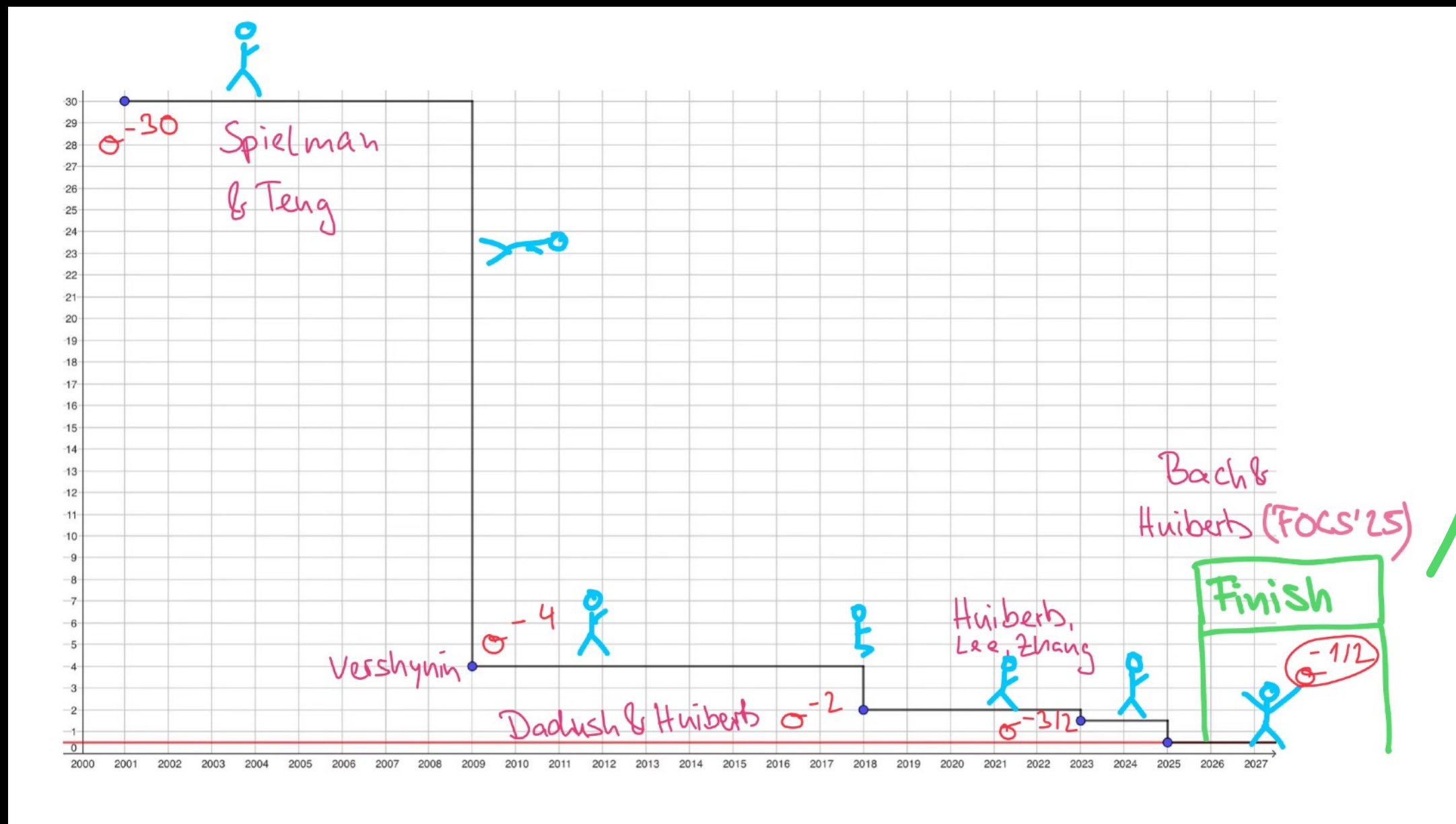




Smoothed analysis
of the Simplex method
(FOCS' 25)

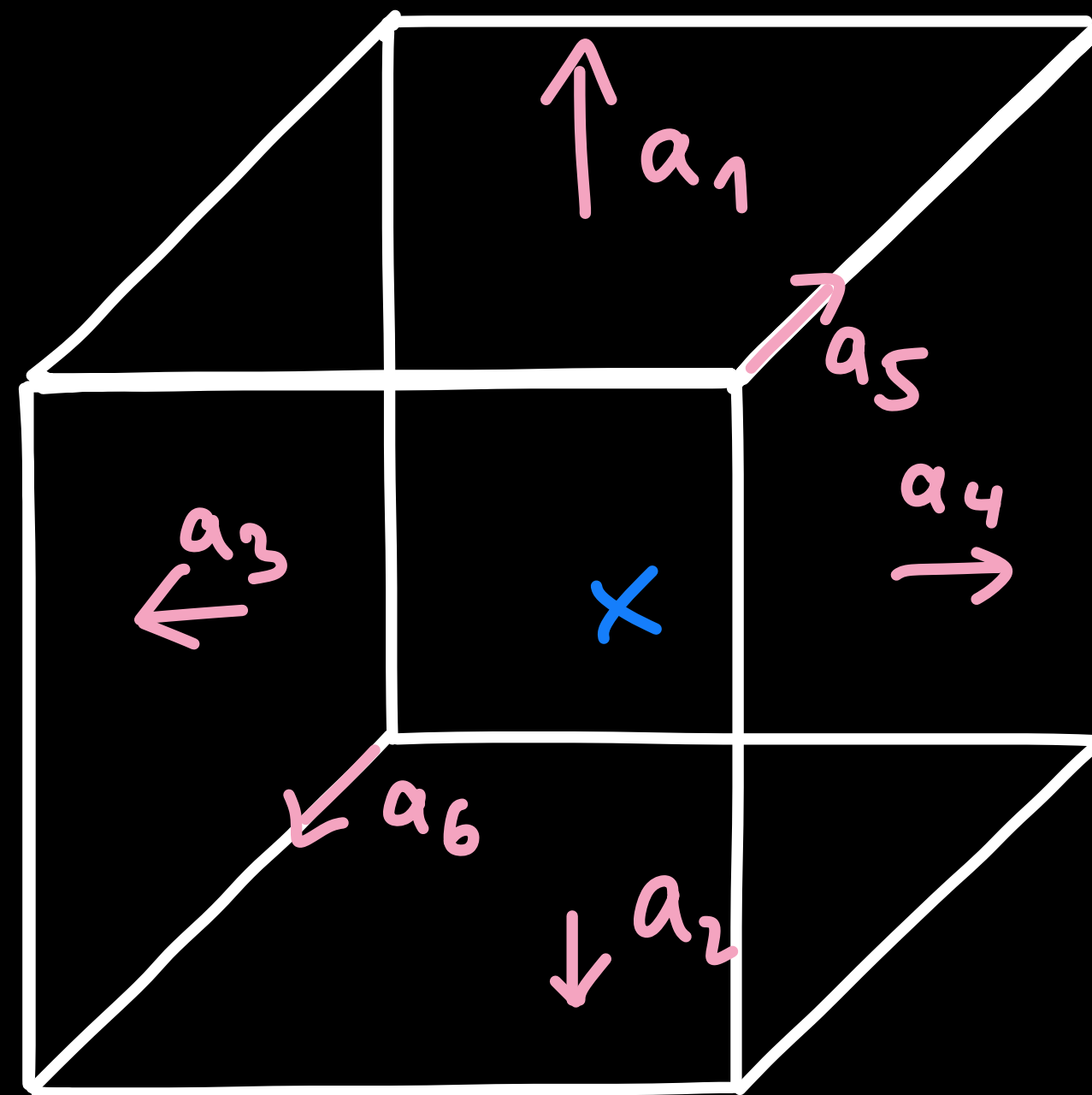
Eleon Bach

based on joint work with Sophie Huiberts

maximize $c^T x$

subject to $Ax \leq b$

$$A = \begin{pmatrix} -a_1 & - \\ \vdots & \\ -a_n & - \end{pmatrix}$$





38. Mathematical Tables Project computers with adding machines

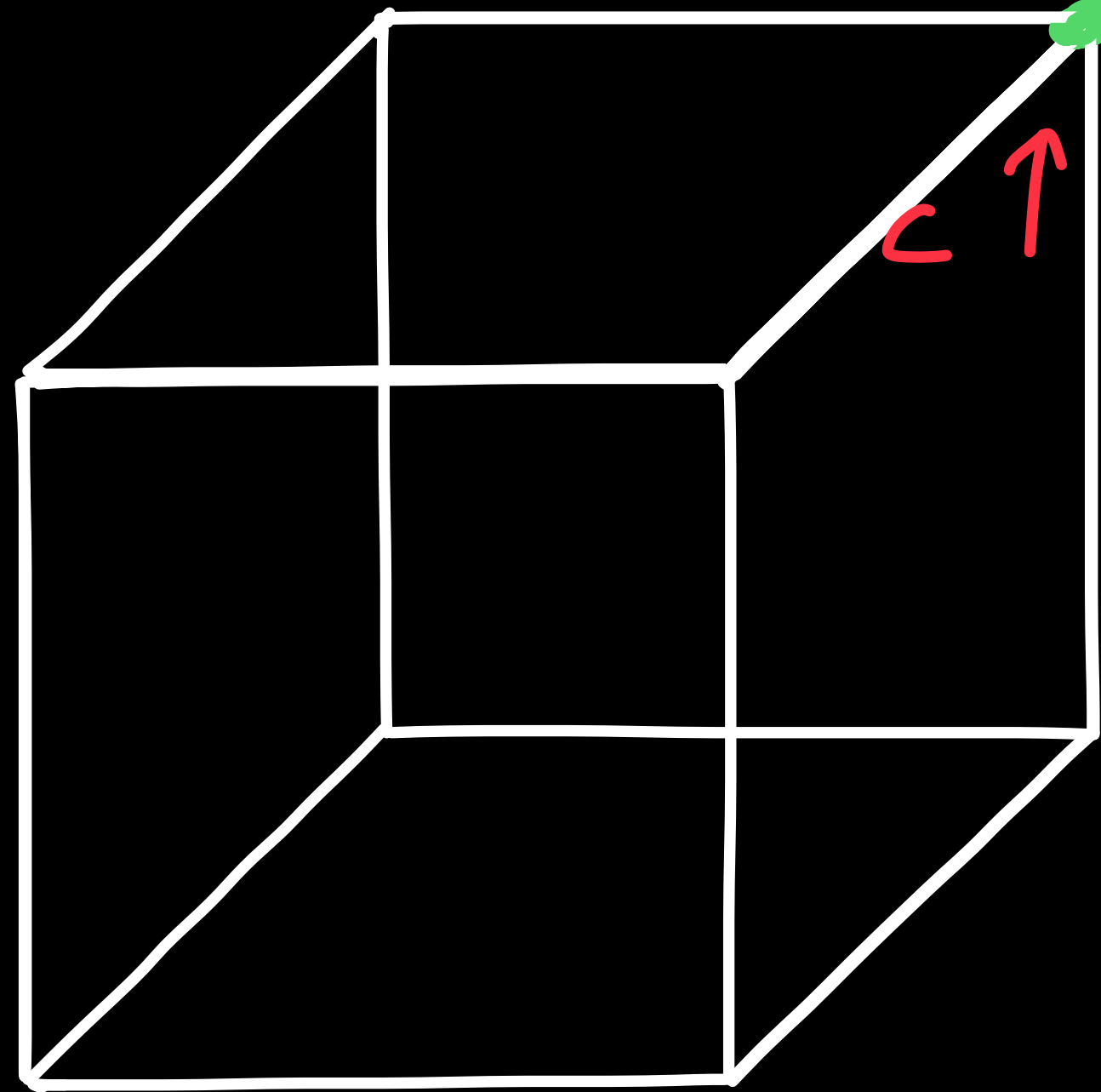
David A. Grier, "When computers were human."

maximize $c^T x$

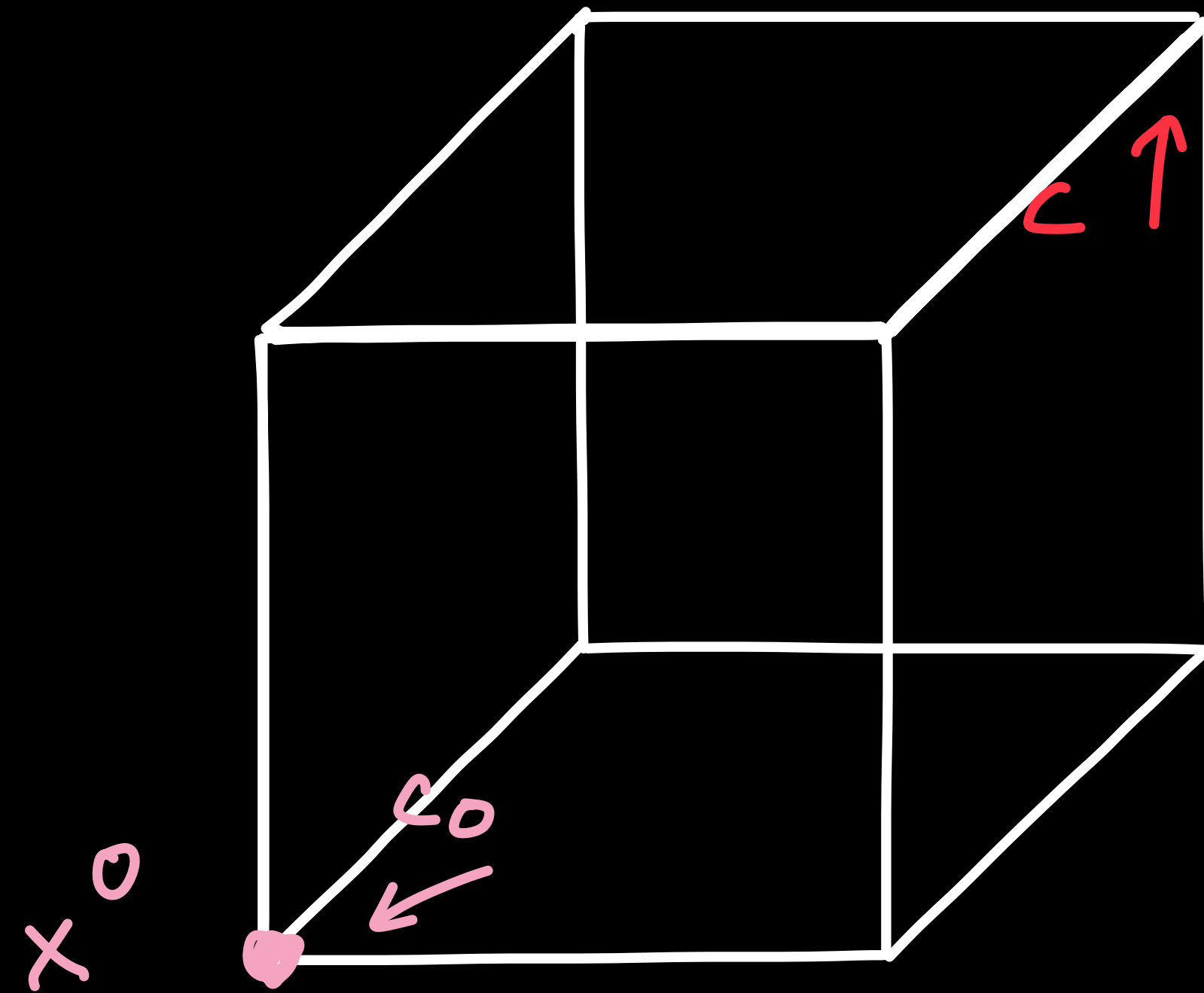
subject to $Ax \leq b$

$$A = \begin{pmatrix} -a_1 & - \\ \vdots & \\ -a_n & - \end{pmatrix}$$

$$\text{opt} = \max_{\{x: Ax \leq b\}} c^T x$$

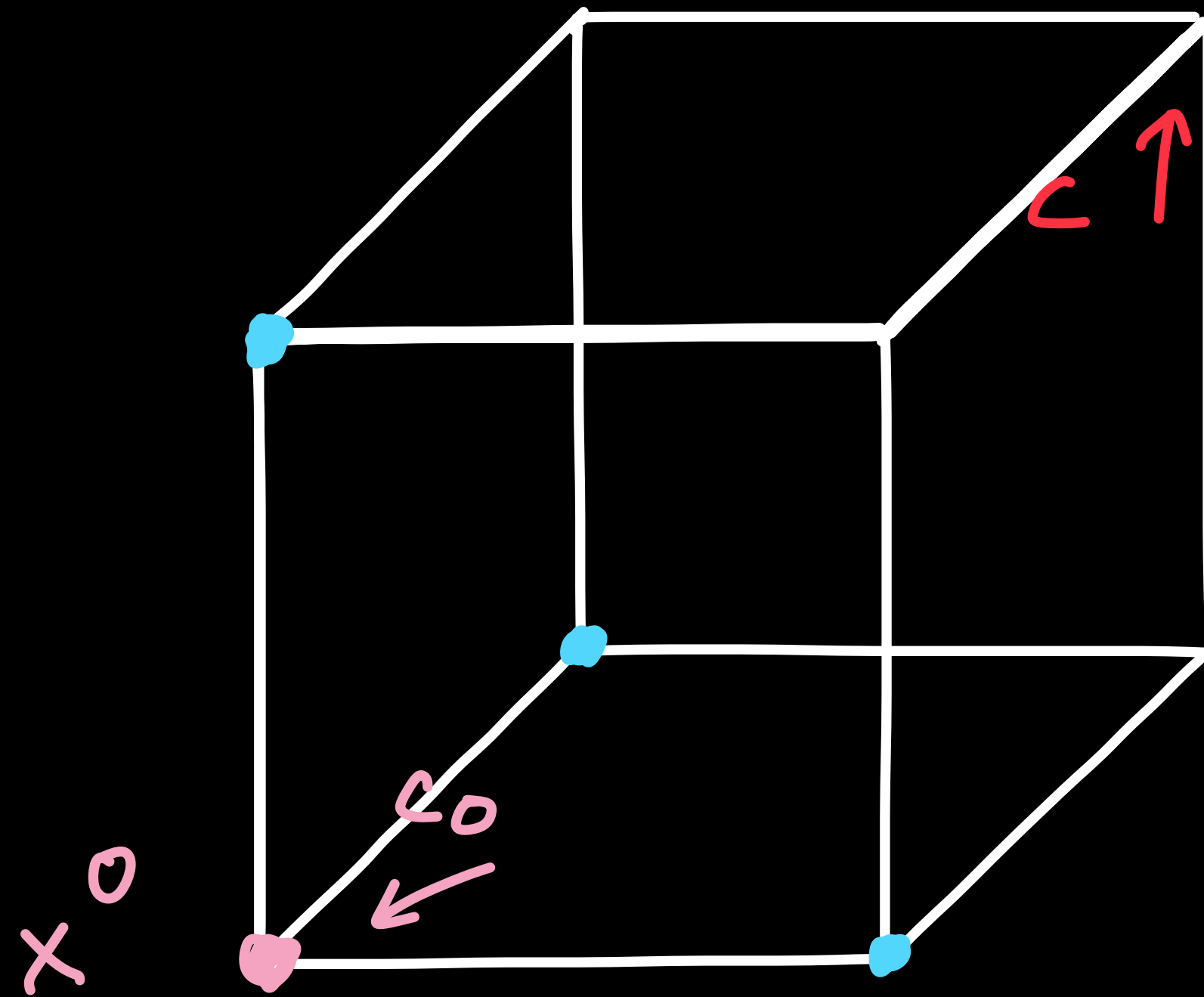


Simplex method



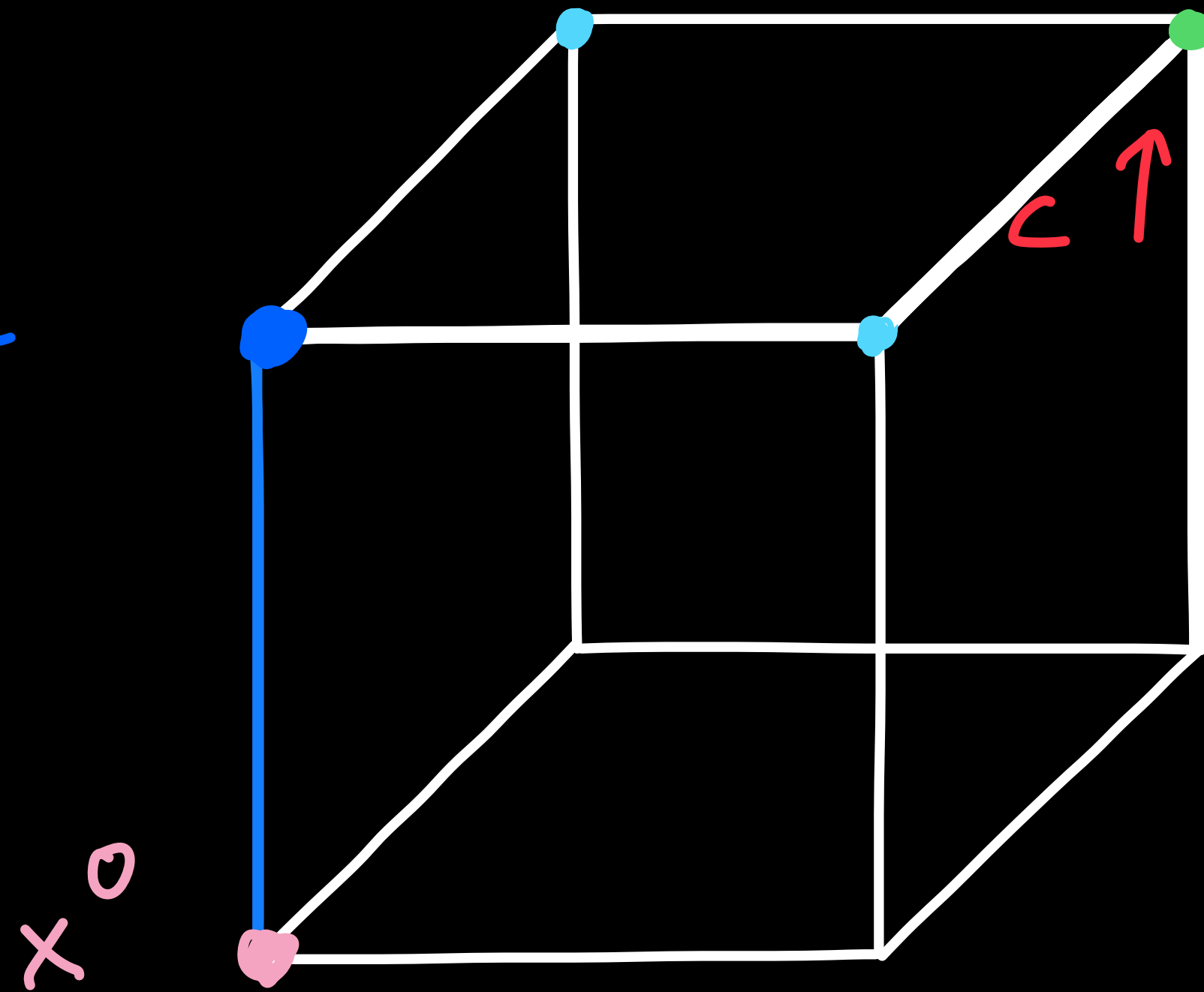
Phase 1

Simplex method



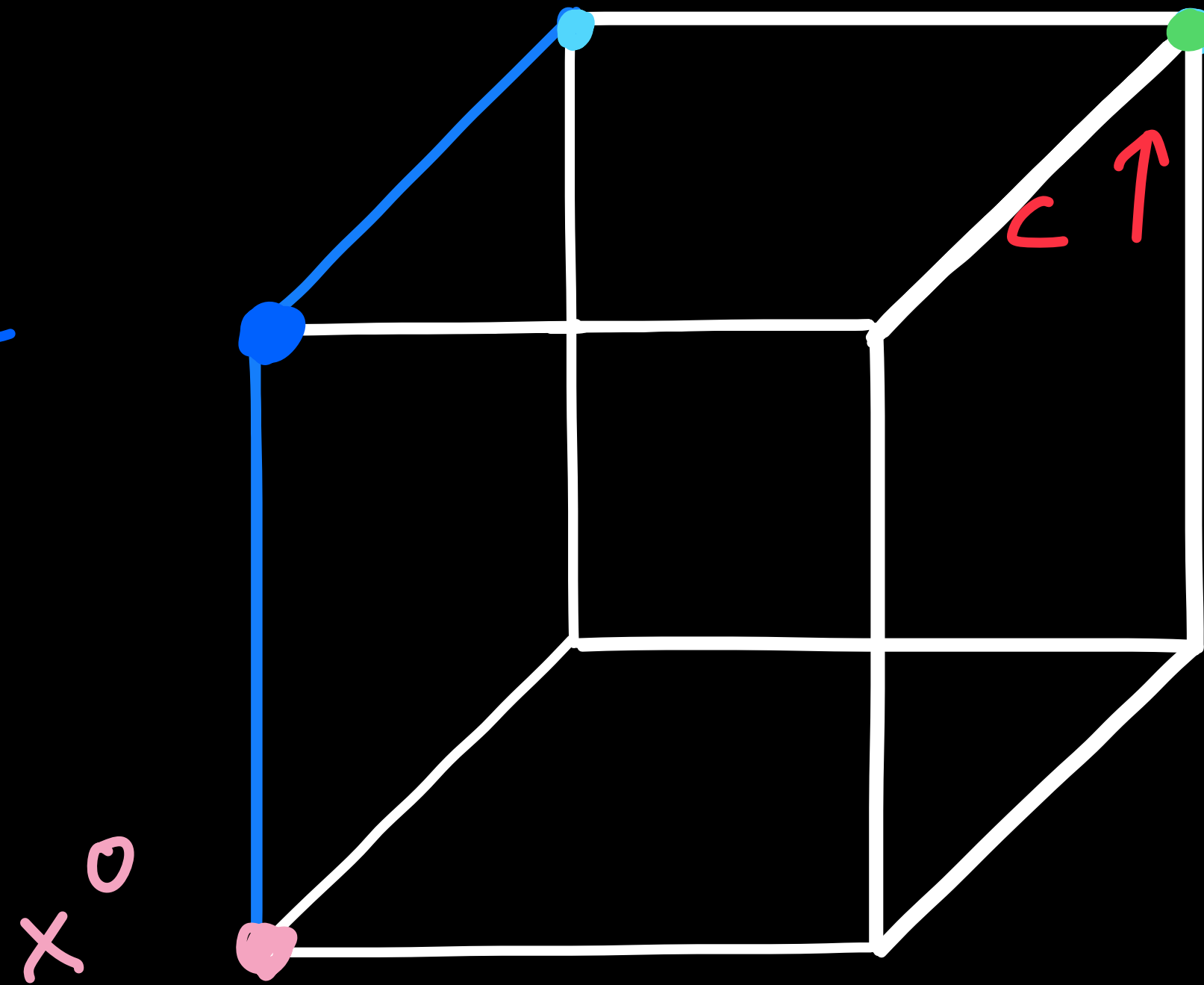
Simplex method

pivot rule



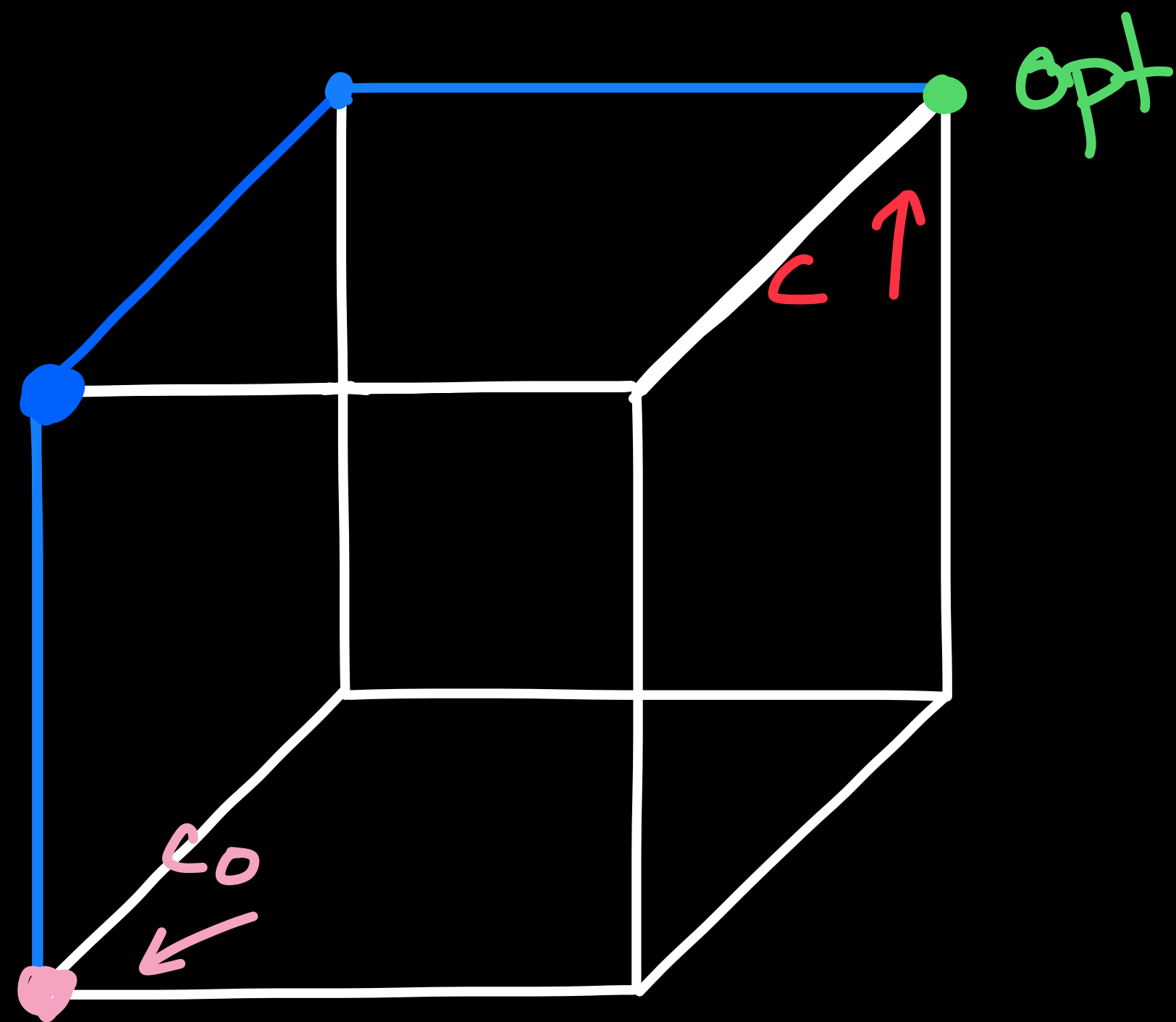
Simplex method

pivot rule



Simplex method

pivot rule



Worst-case complexity

Theorem: The simplex method has
exponential worst-case
complexity.

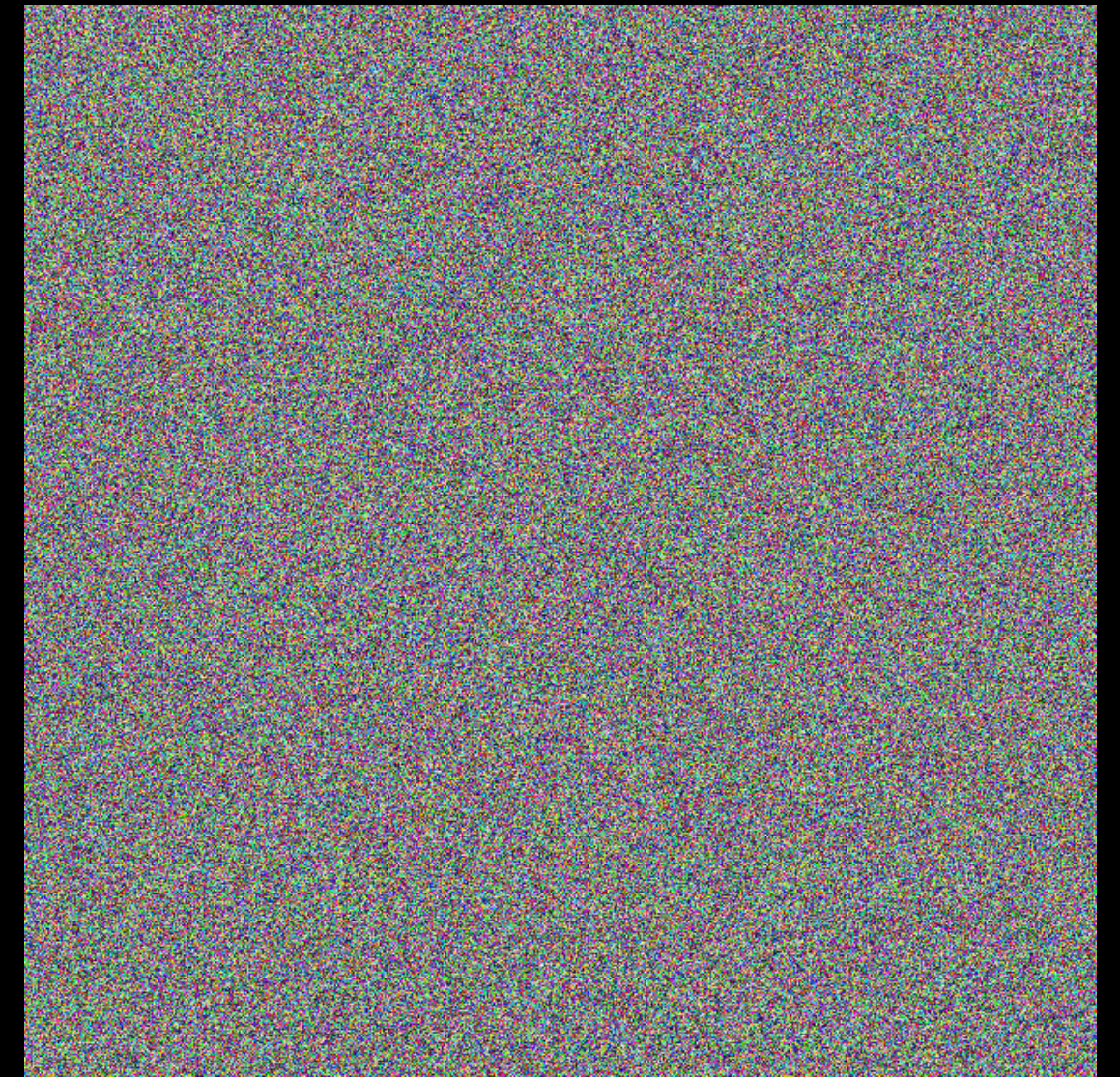
Practice: run time linear in n and d .

How to explain / model this?

Worst case:



Average case:



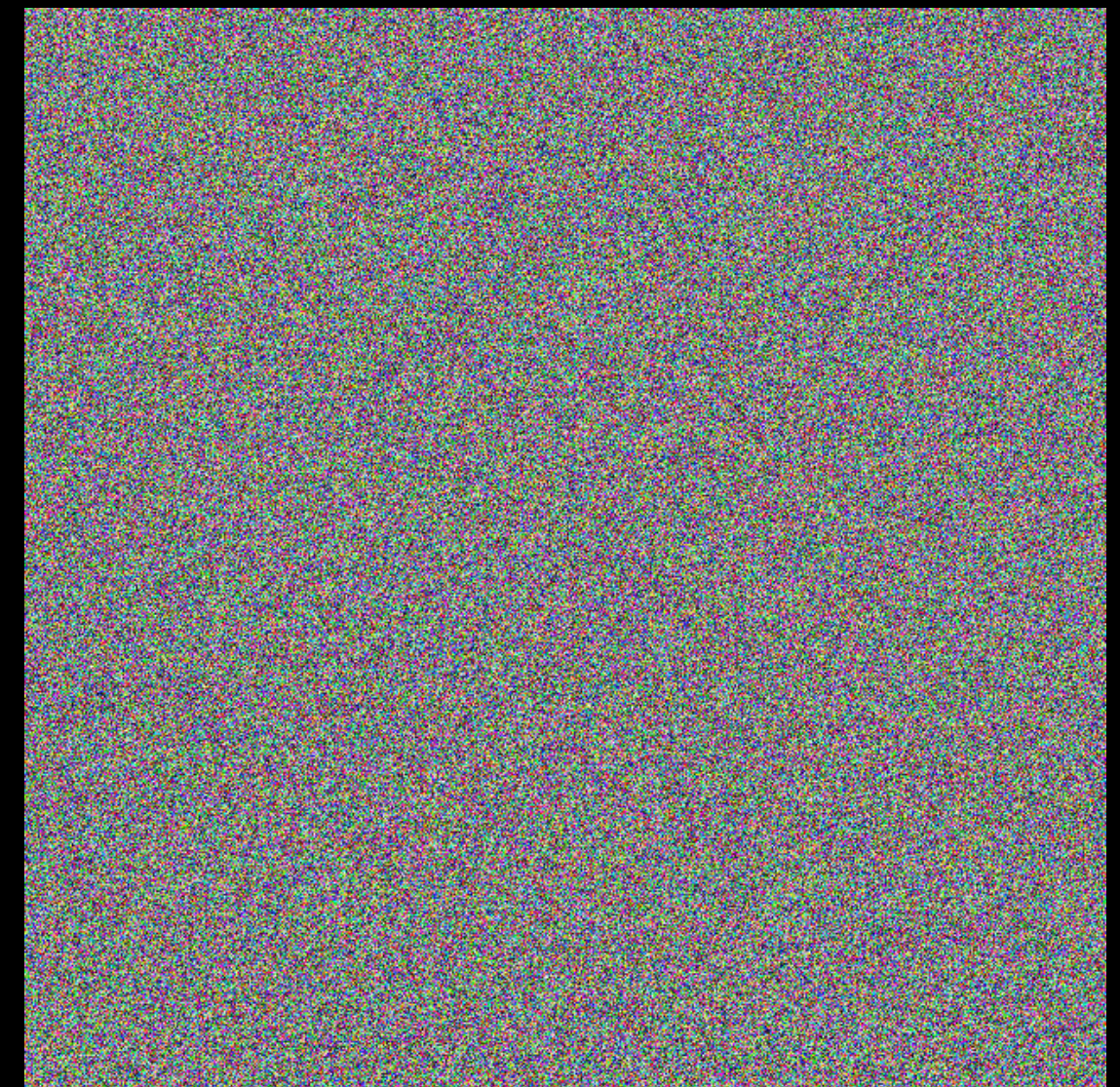
Worst case:
Structure ✓

Upper bounds
(poly) ✗



Average case:
Structure ✗

Upper bounds
(poly) ✓



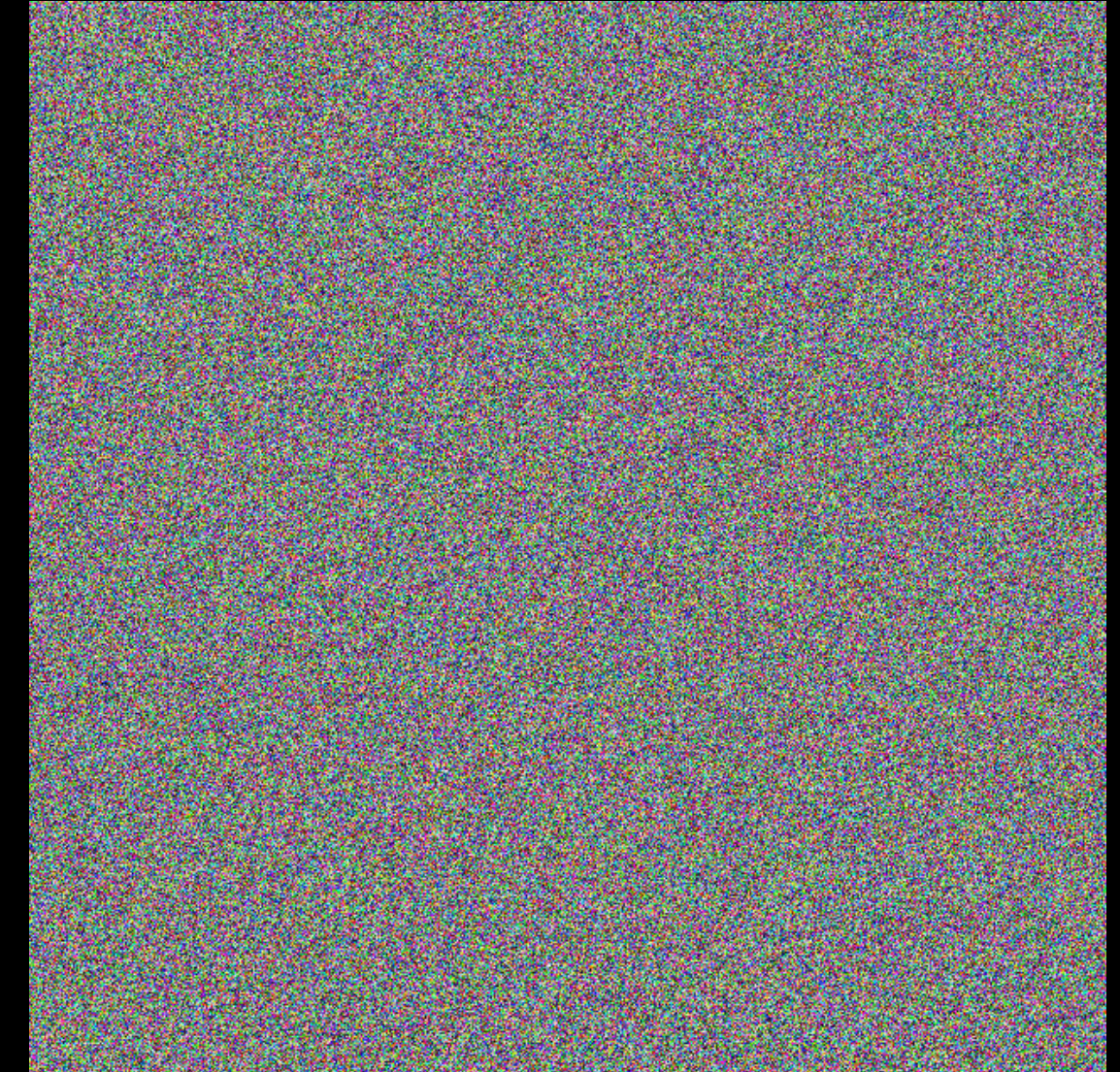
Worst case:
Structure ✓

Upper bounds
(poly) ✗



Average case:
Structure ✗

Upper bounds
(poly) ✓



Worst case:

Structure ✓

Upper bounds
(poly) ✗



Smoothed analysis.

Structure ✓

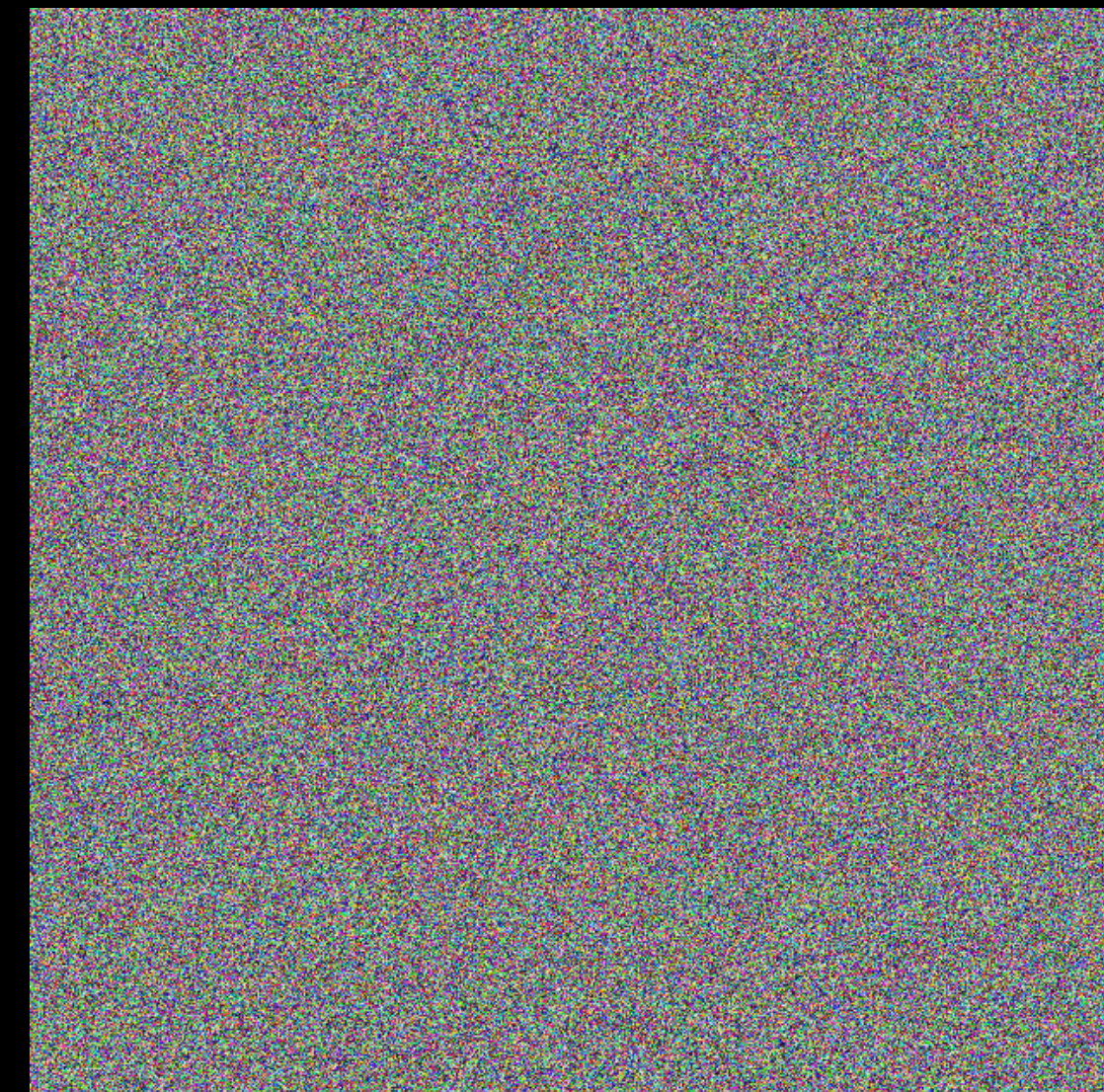
Upper bounds (poly) ✓



Average case:

Structure ✗

Upper bounds
(poly) ✓



Smoothed analysis

Let $\bar{A} \in \mathbb{R}^{n \times d}$ have rows of norm ≤ 1 .

$\bar{b} \in [-1, 1]^n$, $c \in \mathbb{R}^d$

Let $\hat{A}, \hat{b}$ have i.i.d. $N(0, \sigma^2)$ entries

Smoothed analysis

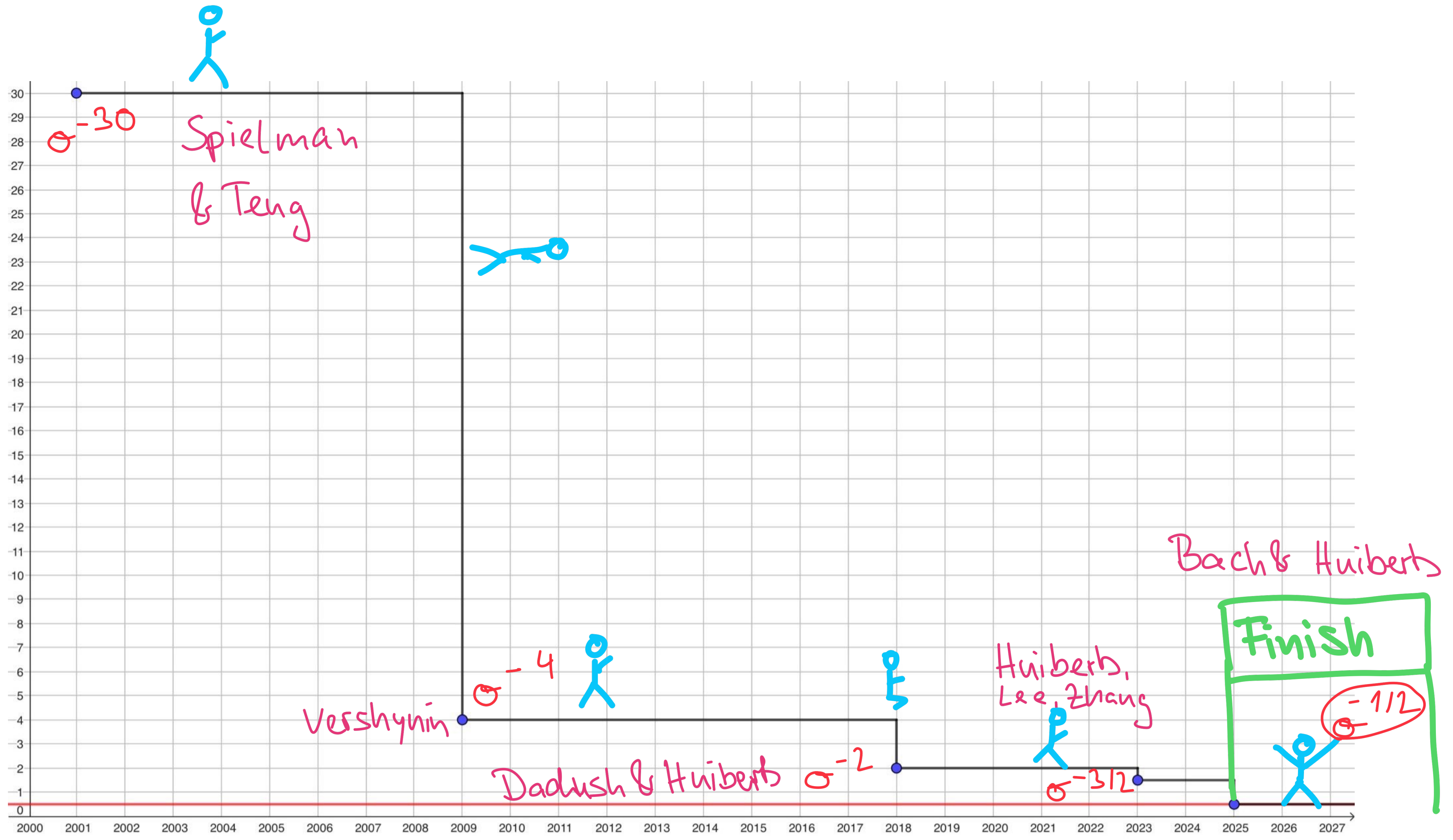
Let $\bar{A} \in \mathbb{R}^{n \times d}$ have rows of norm ≤ 1 .

$$\bar{b} \in [-1, 1]^n, c \in \mathbb{R}^d$$

Let $\hat{A}, \hat{b}$ have i.i.d. $N(0, \sigma^2)$ entries

Theorem:

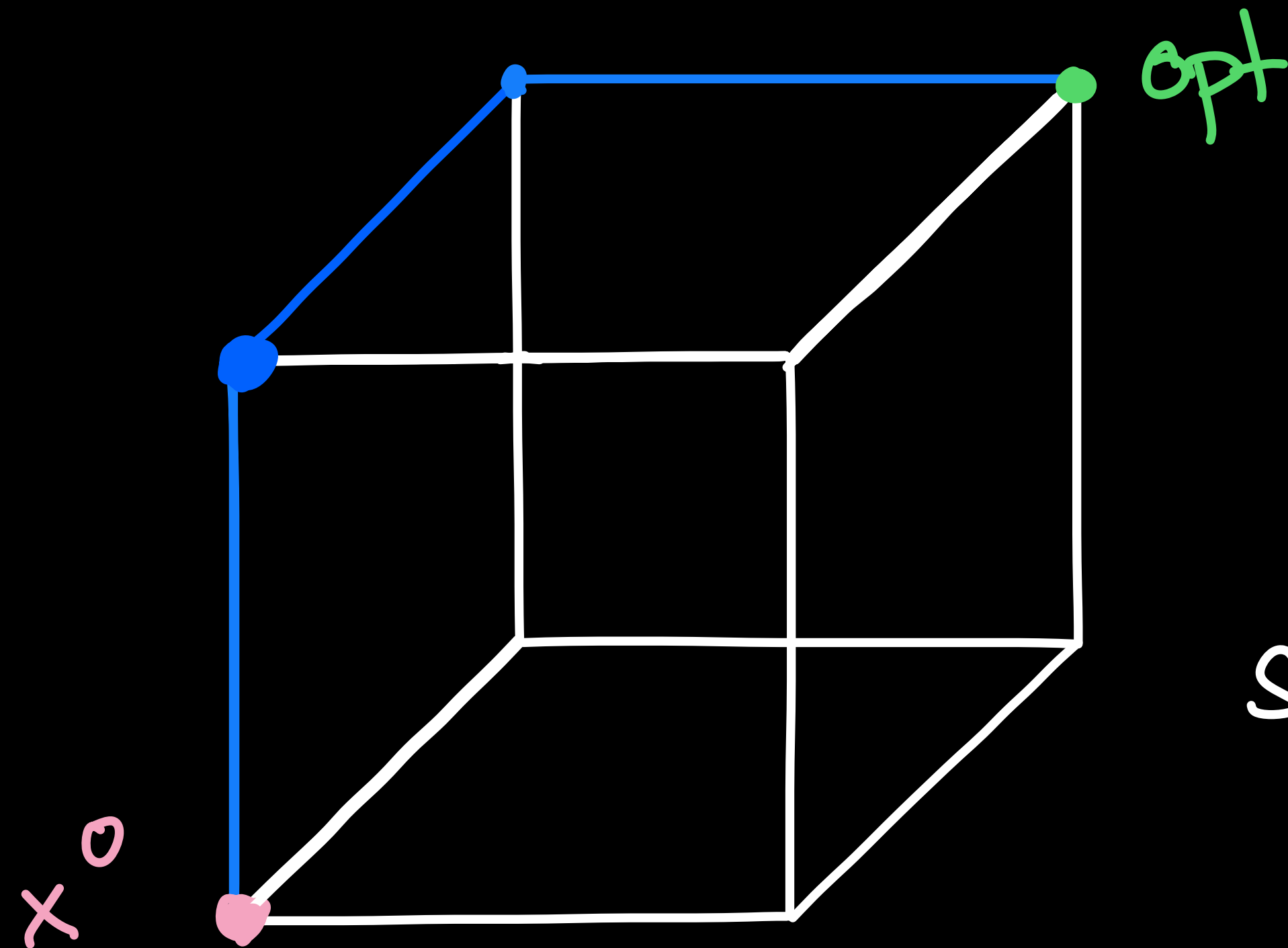
$$\max_{\bar{A}, \bar{b}, c} \mathbb{E}_{\hat{A}, \hat{b}} \left[\begin{array}{l} \text{time to solve} \\ \max c^T x \\ \text{s.t. } (\bar{A} + \hat{A})x \leq \bar{b} + \hat{b} \end{array} \right] = \sigma^{-\frac{1}{2}} \cdot \text{poly}(d, \log n)$$



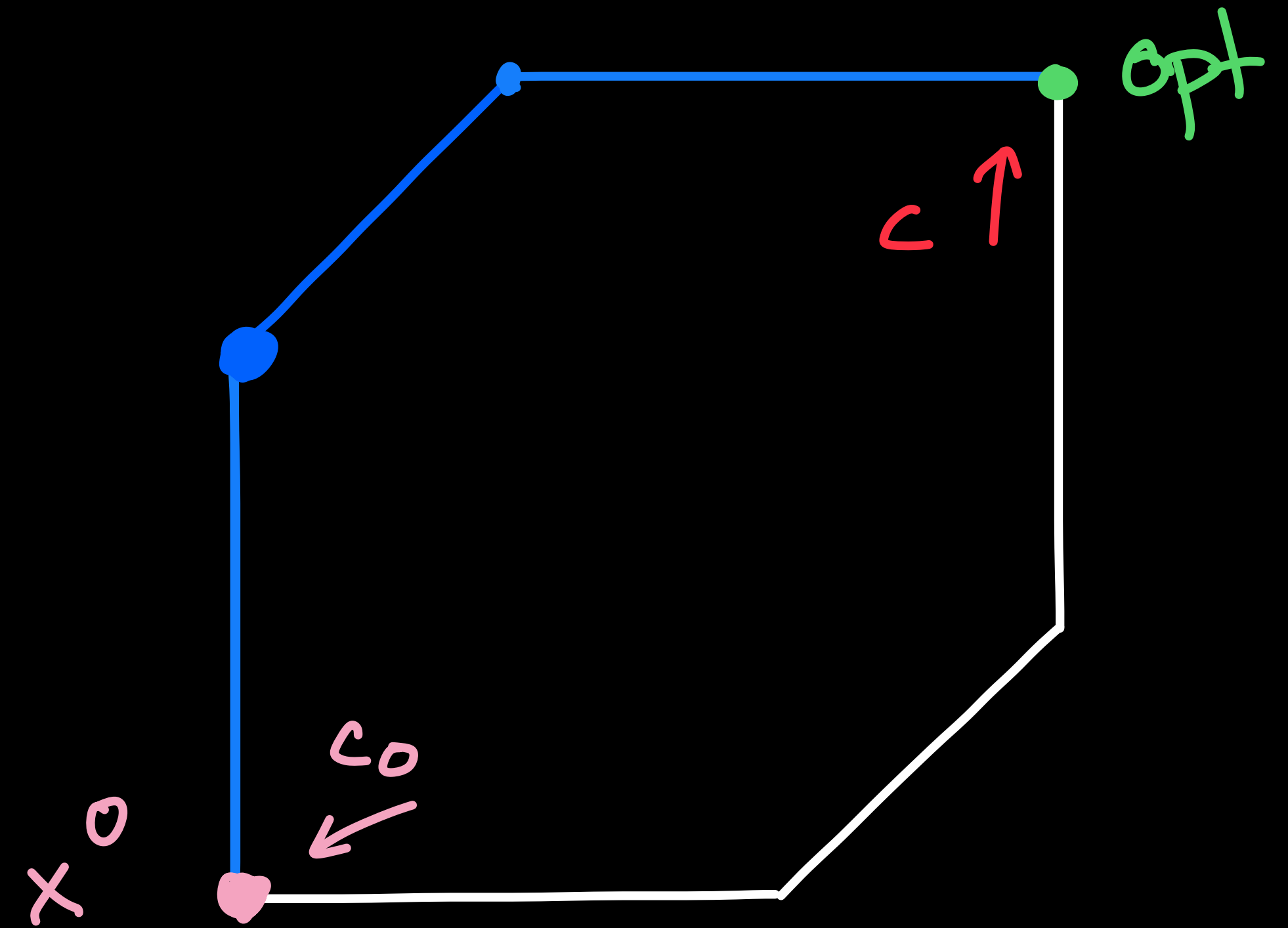
Shadow vertex

Simplex

method



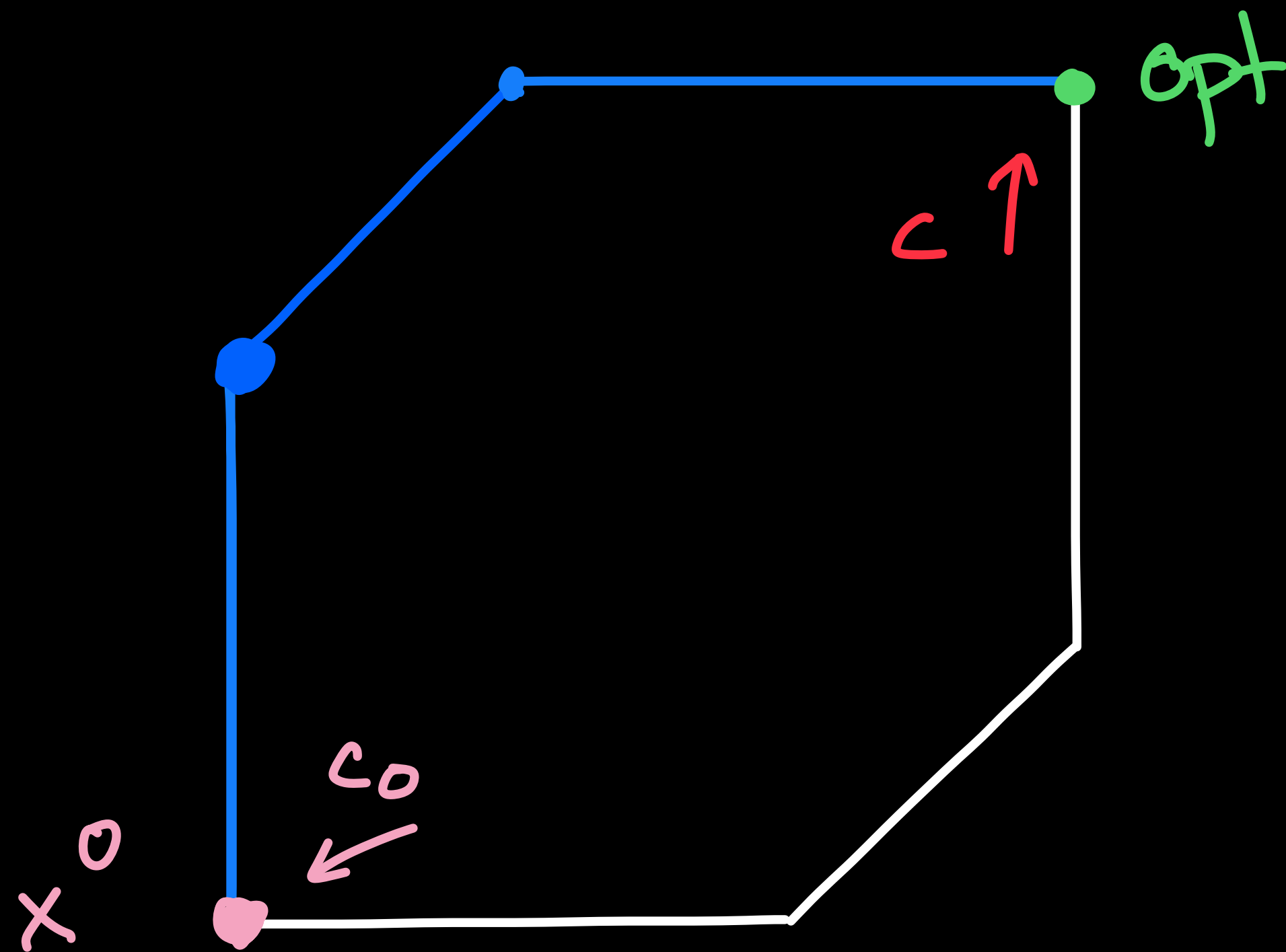
project to
→
 $\text{span}(c_0, c)$



Shadow vertex

Simplex

method



- solutions to

$$\max (\lambda c_0 + (1-\lambda)c)^T x$$

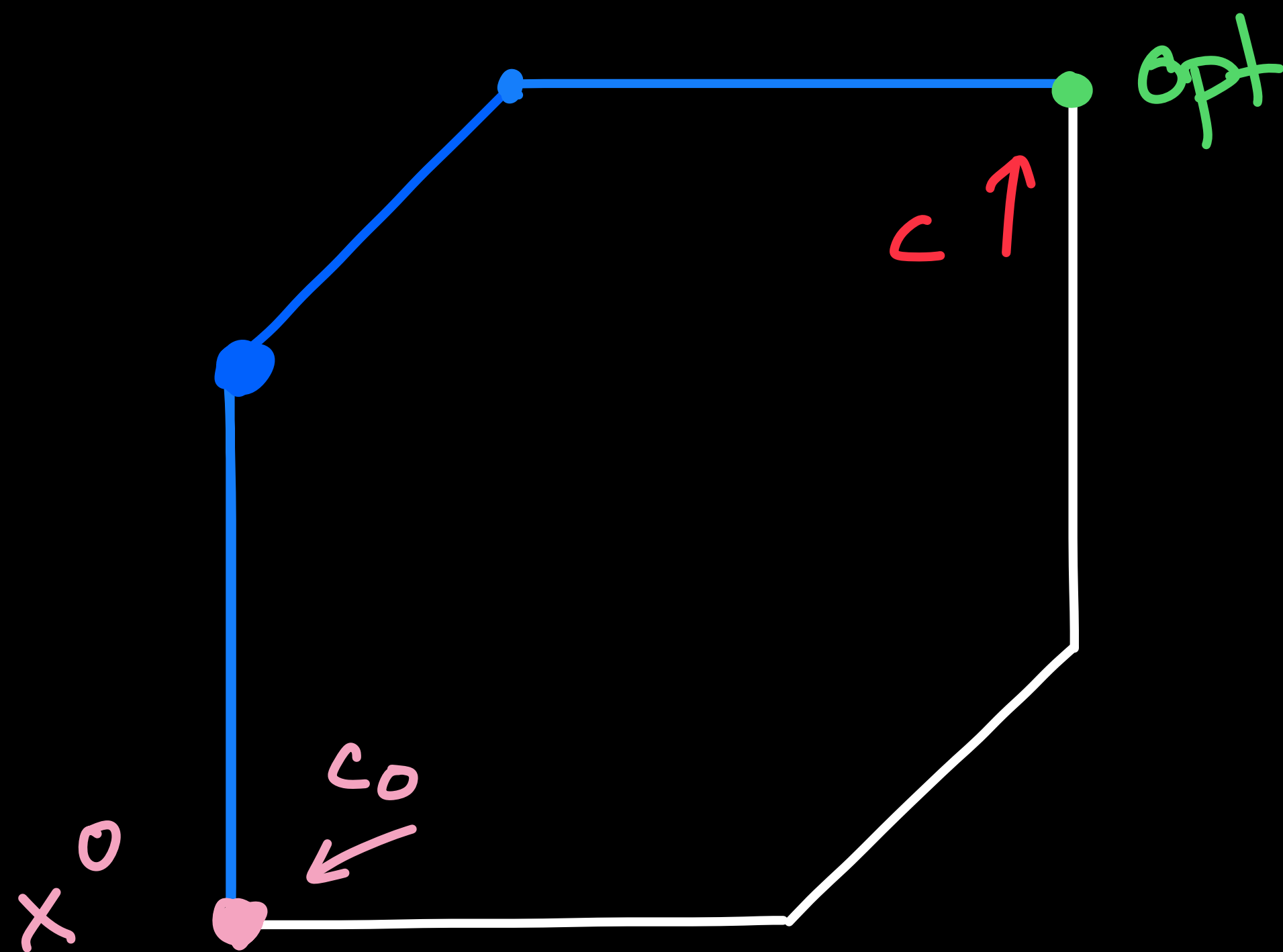
$$\text{s.t. } Ax \leq b$$

form simplex path

Shadow vertex

Simplex

method



- solutions to

$$\max (\lambda c_0 + (1-\lambda)c)^T x$$

$$\text{s.t. } Ax \leq b$$

form simplex path

→ we bound # vertices on path

Our improvements

1. pick c_0 random, c fixed (apart algo reduction)

Our improvements

1. pick c_0 random, c fixed (apart algo reduction)
2. pick two clever potentials
"perimeter integral" & "exterior angle"

Our improvements

1. pick c_0 random, c fixed (apart algo reduction)

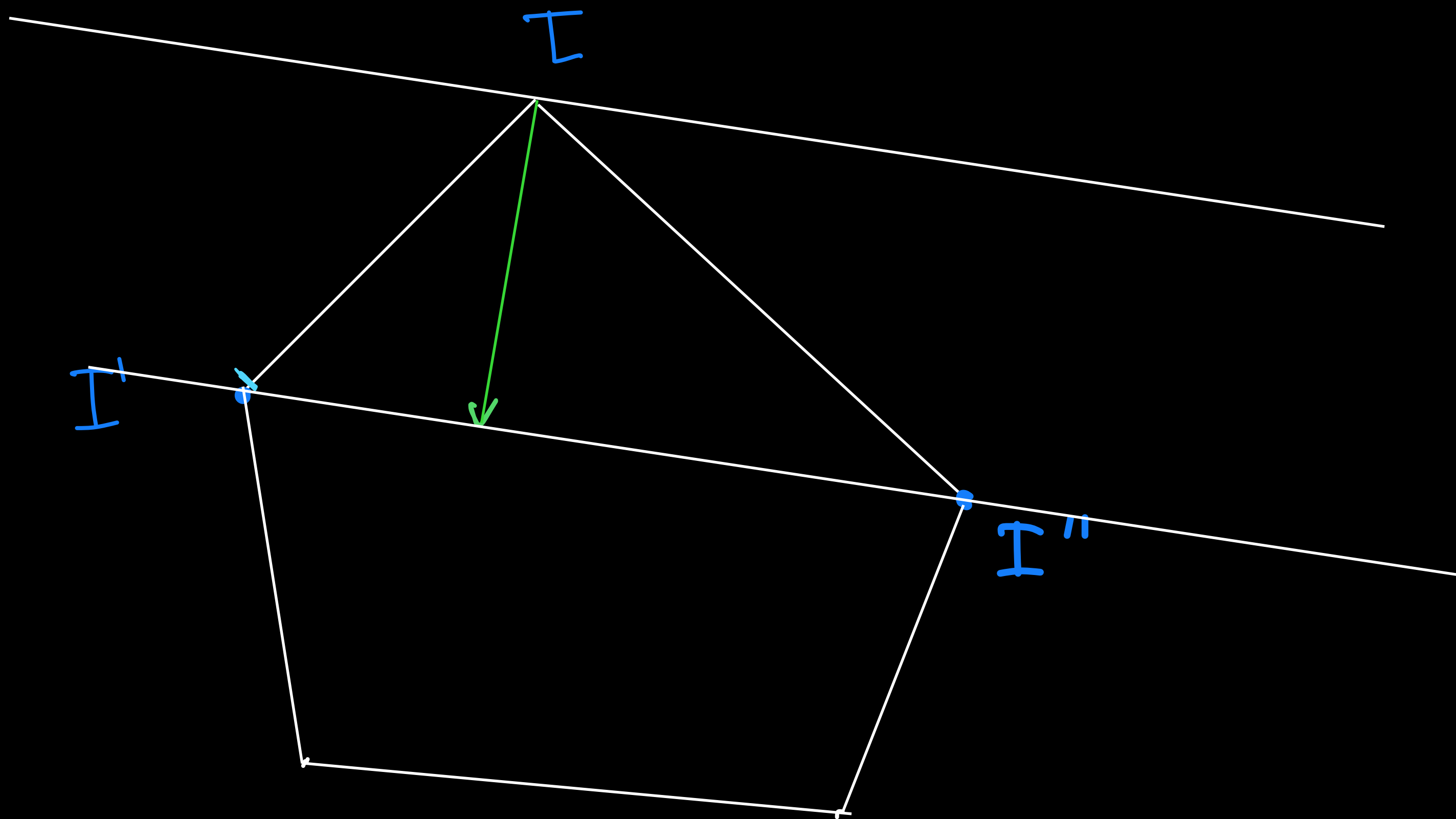
2. pick two clever potentials

"perimeter integral" & "exterior angle"

3. combine them to vertex separation

→ each step consumes a lot of either integral

Want: Each vertex of
projection is
well-separated
from the others



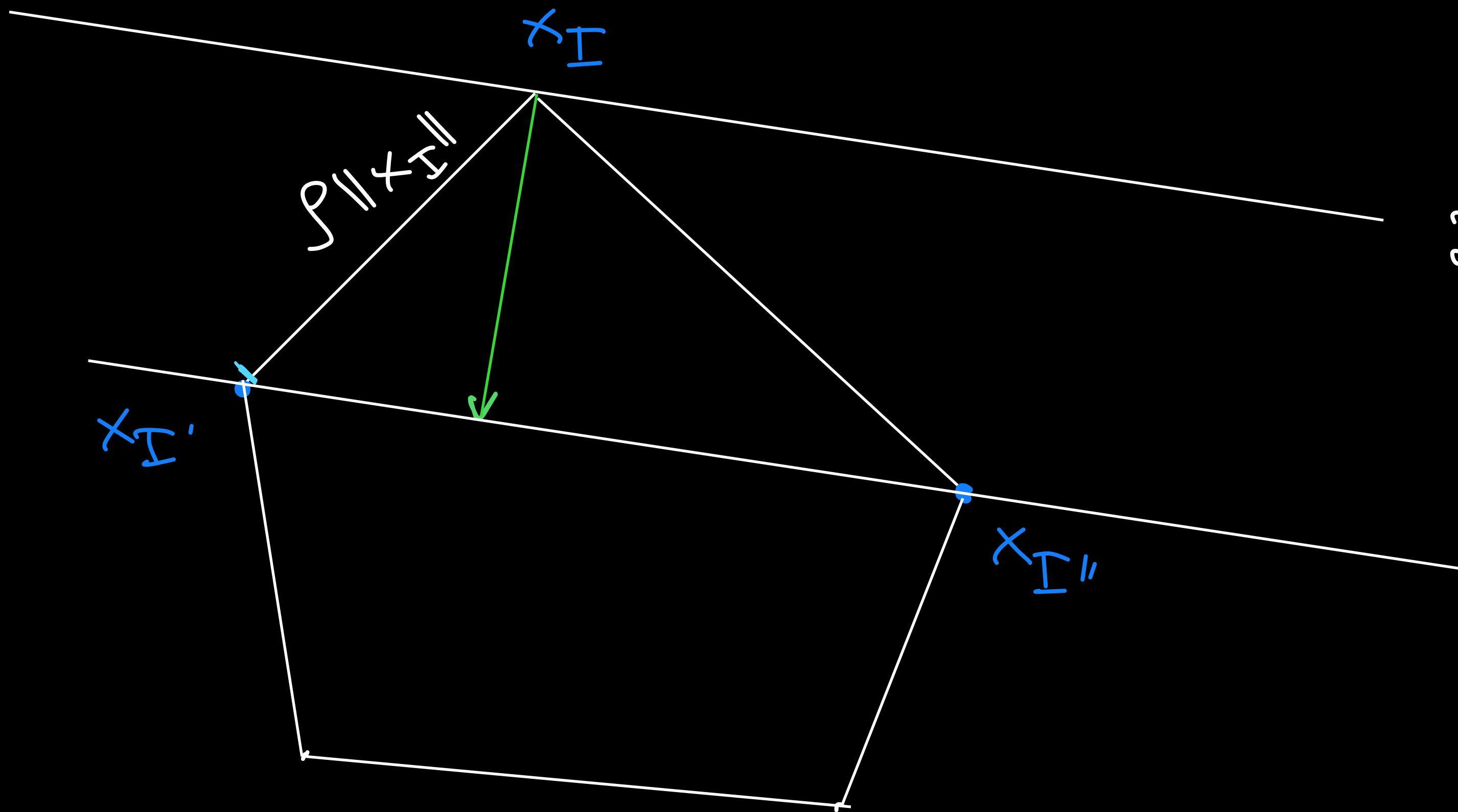
$$1) \|x_I - x_{I'}\| \leq f \|x_I\|$$

→ large exterior angle

$$2) \|x_I - x_{I'}\| > \rho \|x_I\|$$

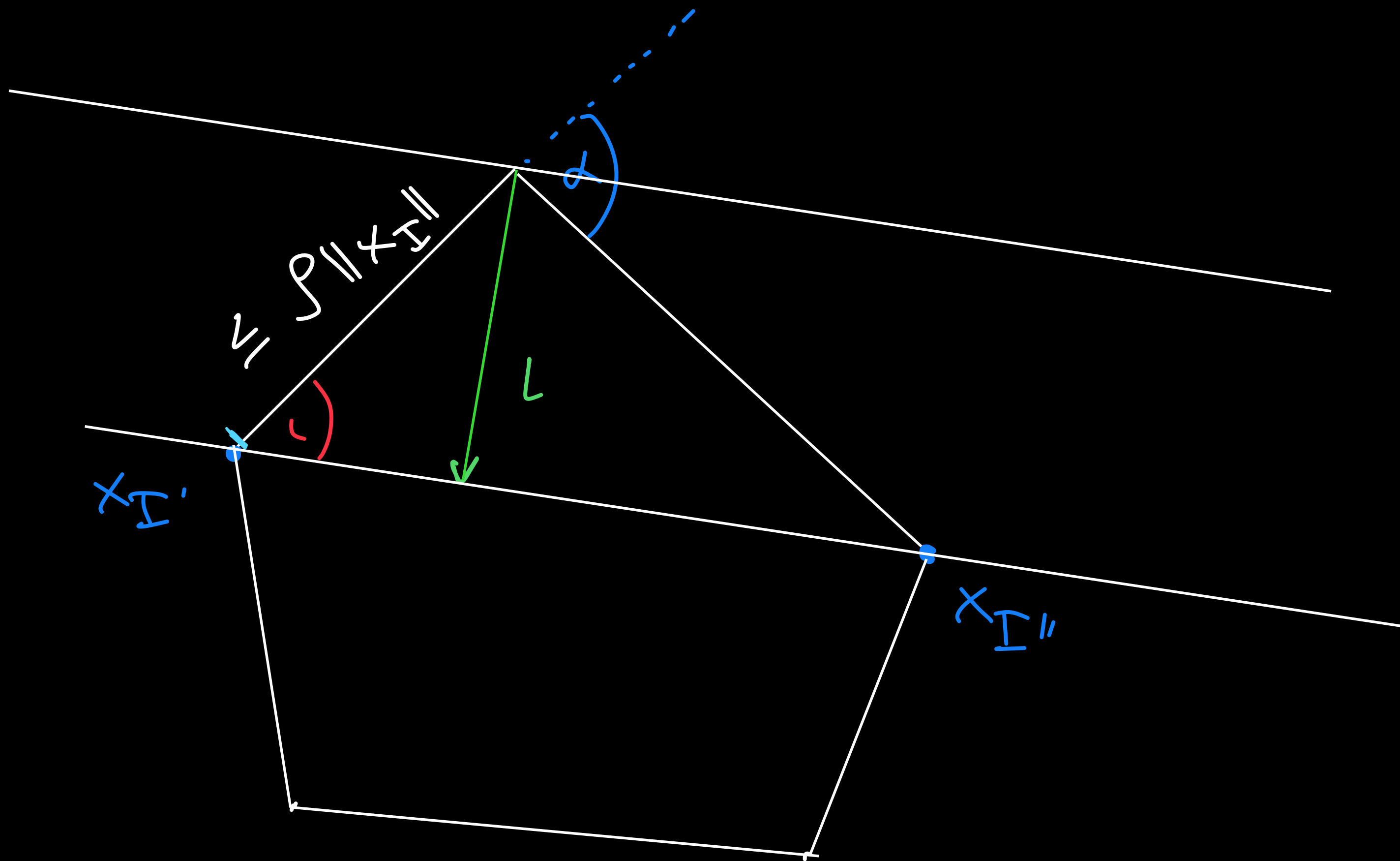
→ big portion of perimeter

$$\text{integral} \int_{\mathbb{O}Q} \|t^{-1}\| dt$$



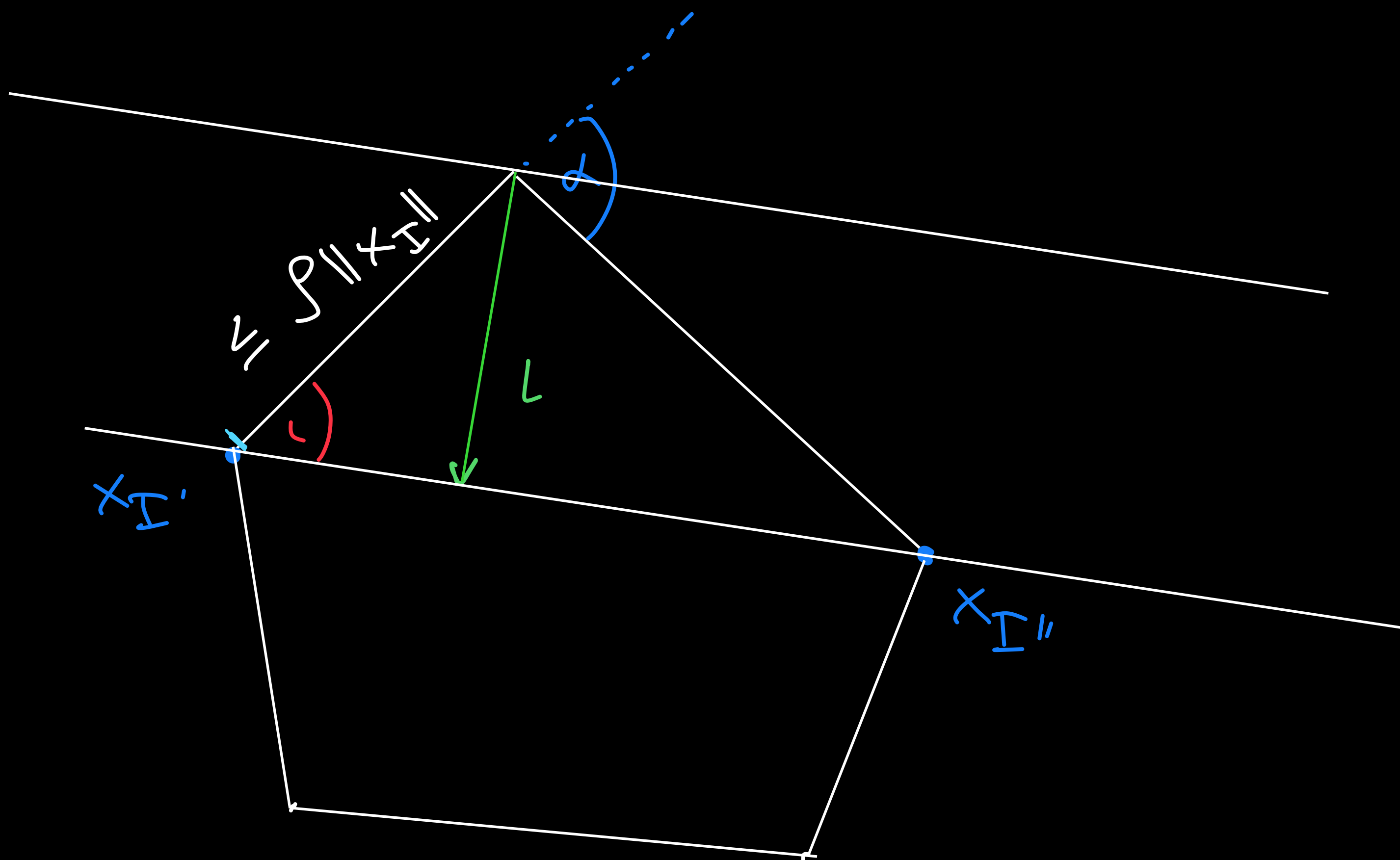
Case 1) $\|x_I - x_{I'}\| \leq f \|x_I\|$

$\alpha \geq$



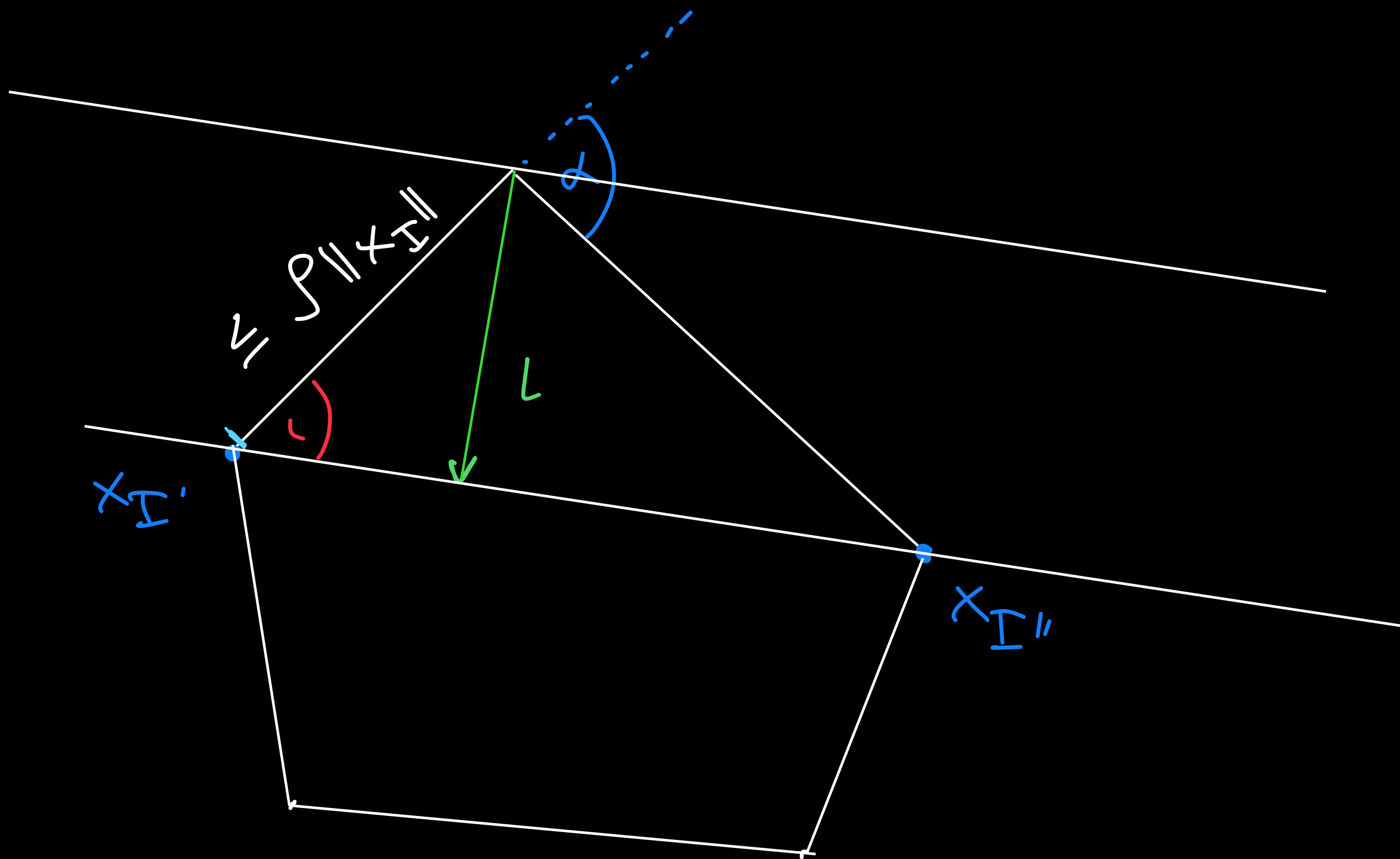
Case 1) $\|x_I - x_{I'}\| \leq f \|x_I\|$

$\alpha \geq \epsilon$

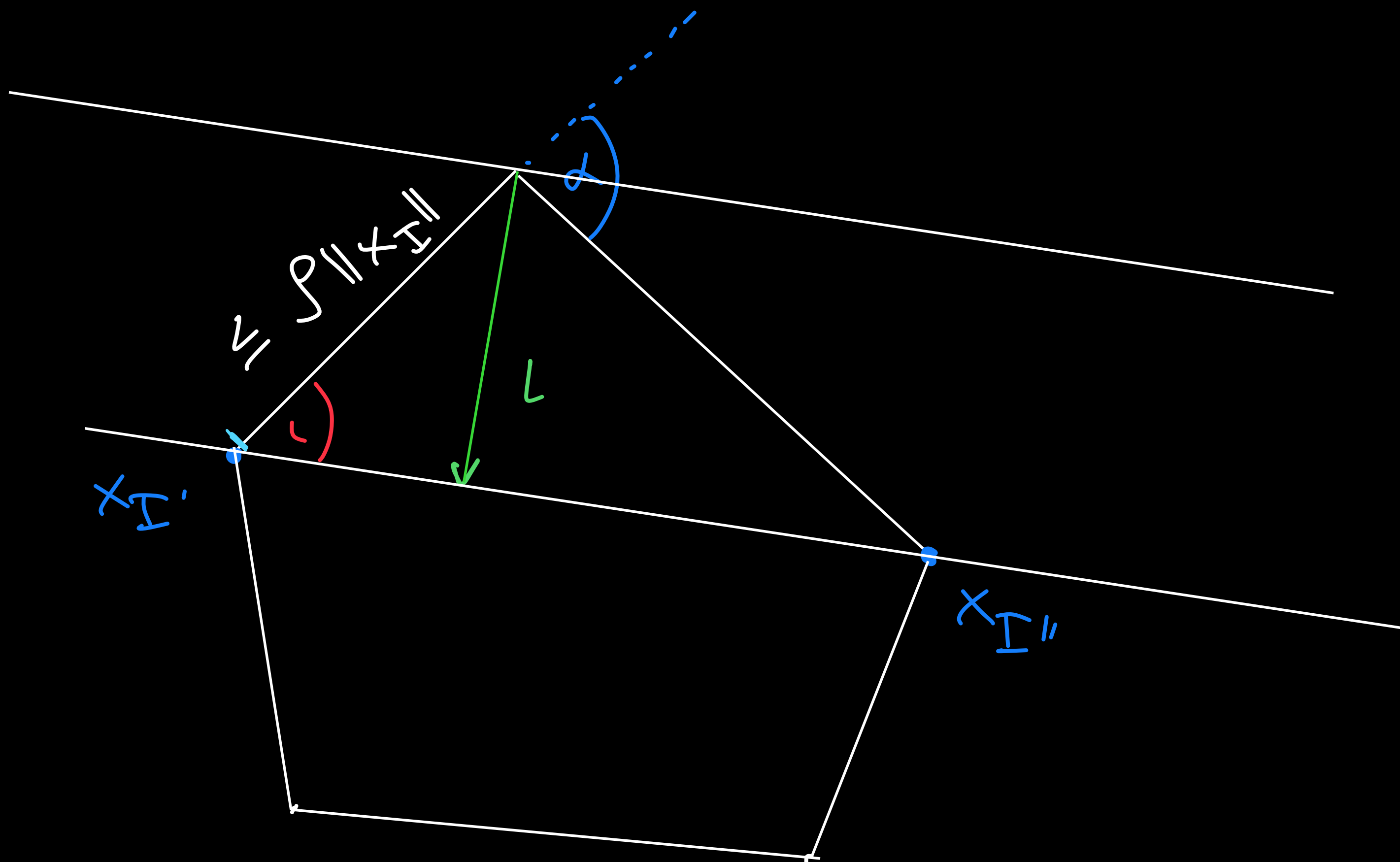


Case 1) $\|x_I - x_{I'}\| \leq f \|x_I\|$

$$\alpha \geq \epsilon \geq \sin(\epsilon)$$



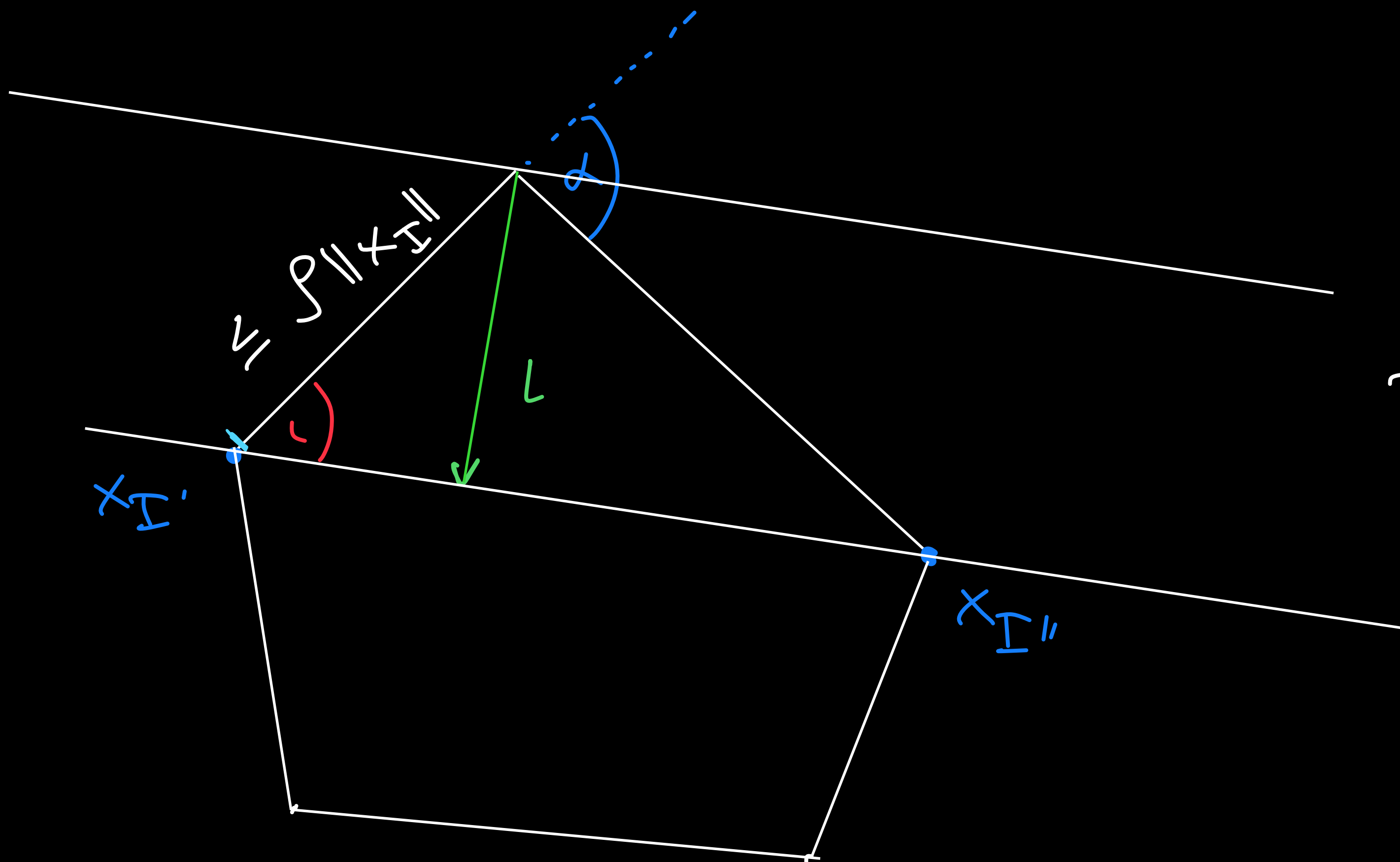
Case 1) $\|x_I - x_{I'}\| \leq f \|x_I\|$



$$\alpha \geq \gamma \geq \sin(\gamma)$$

$$= \frac{L}{f \|x_I\|}$$

Case 1) $\|x_I - x_{I'}\| \leq f \|x_I\|$

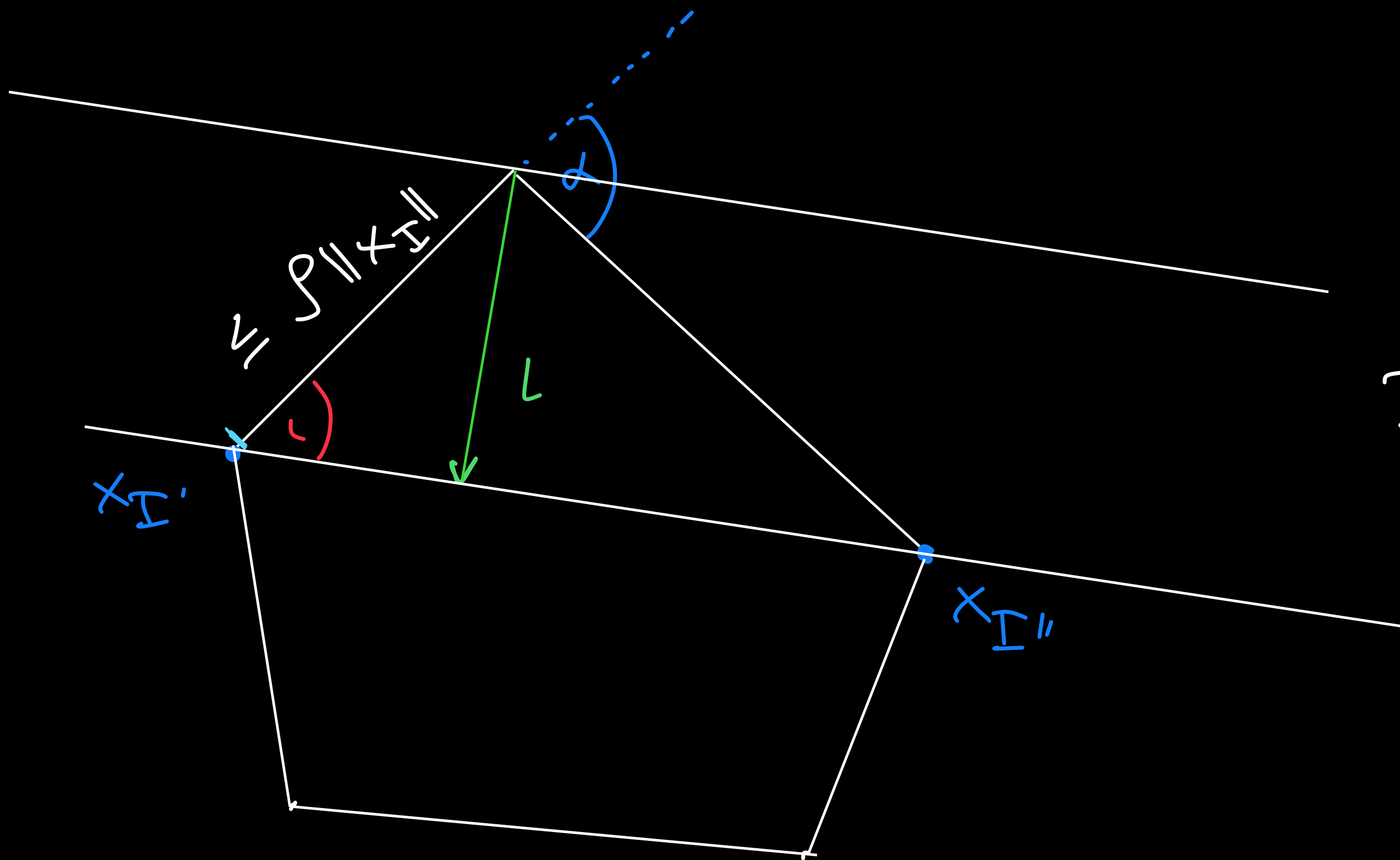


$$\alpha \geq L \geq \sin(\alpha)$$

$$= \frac{L}{f \|x_I\|}$$

Full sum: $\sum \alpha = 2\pi$
 α exterior angle

Case 1) $\|x_I - x_{I'}\| \leq f \|x_I\|$



$$\alpha \geq L \geq \sin(L)$$

$$= \frac{L}{f \|x_I\|}$$

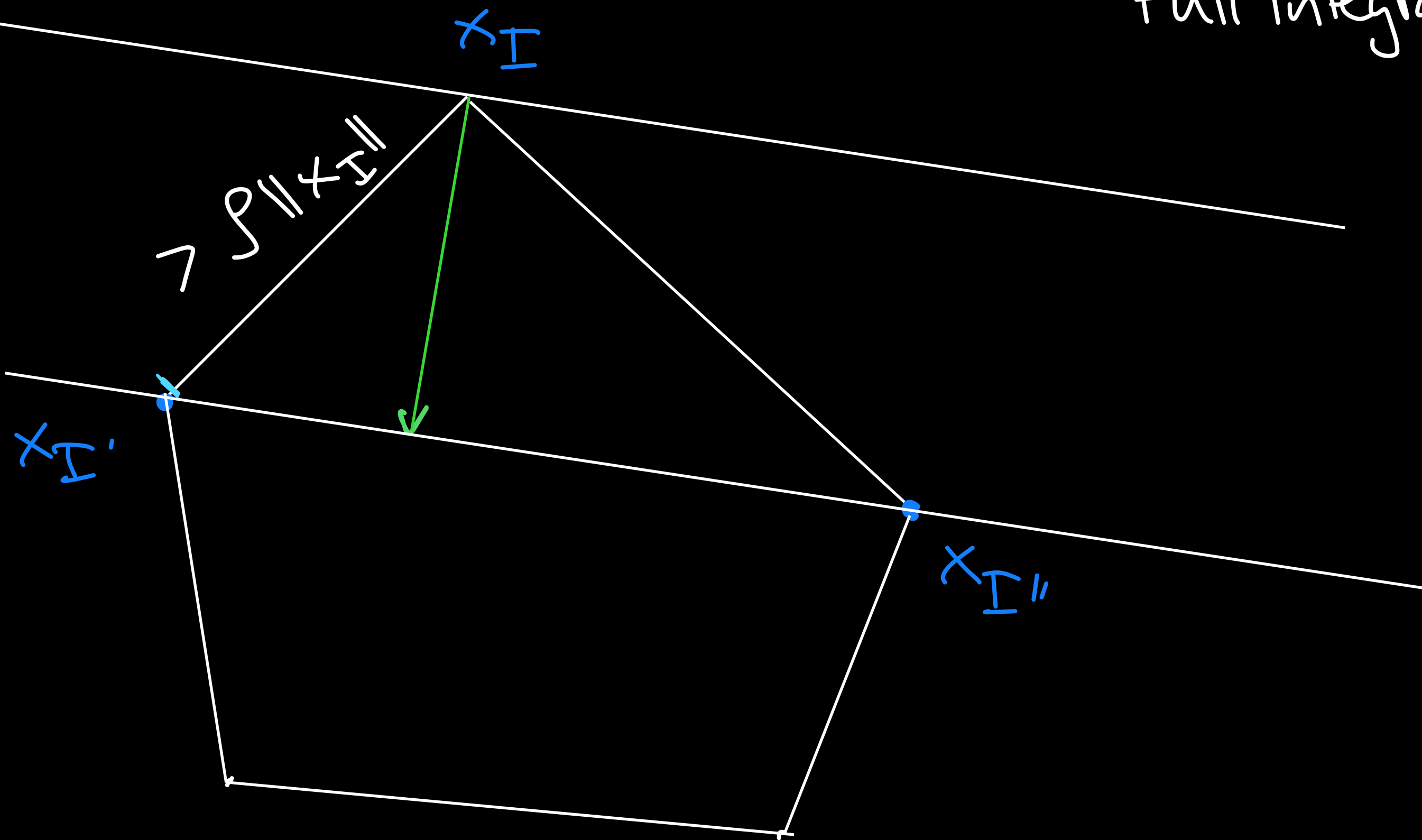
Full sum: $\sum \alpha = 2\pi$
 α exterior angle

$\Rightarrow$ # edges of case 1

$$\leq \frac{2\pi}{f \|x_I\|} \cdot L$$

Case: 2) $\|x_I - x_{I'}\| > \int \|x_I\|$

Full integral: $\int_{\partial Q} \|t\|^{-1} dt \leq O(d \log n)$

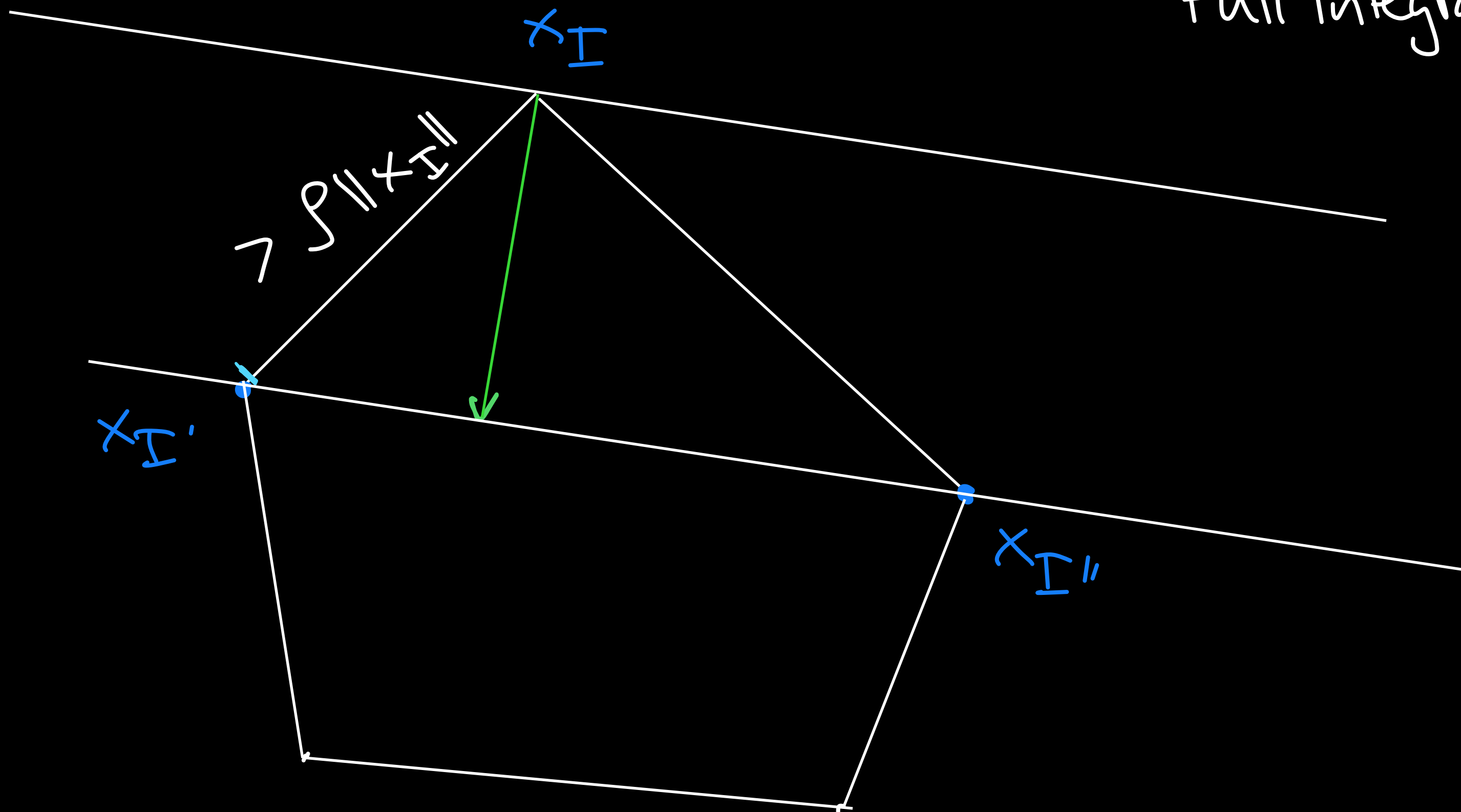


Case: 2) $\|x_I - x_{I'}\| > f \|x_I\|$

Full integral: $\int_{\partial Q} \|t\|^{-1} dt \leq O(d \log n)$

edge long $\Rightarrow$

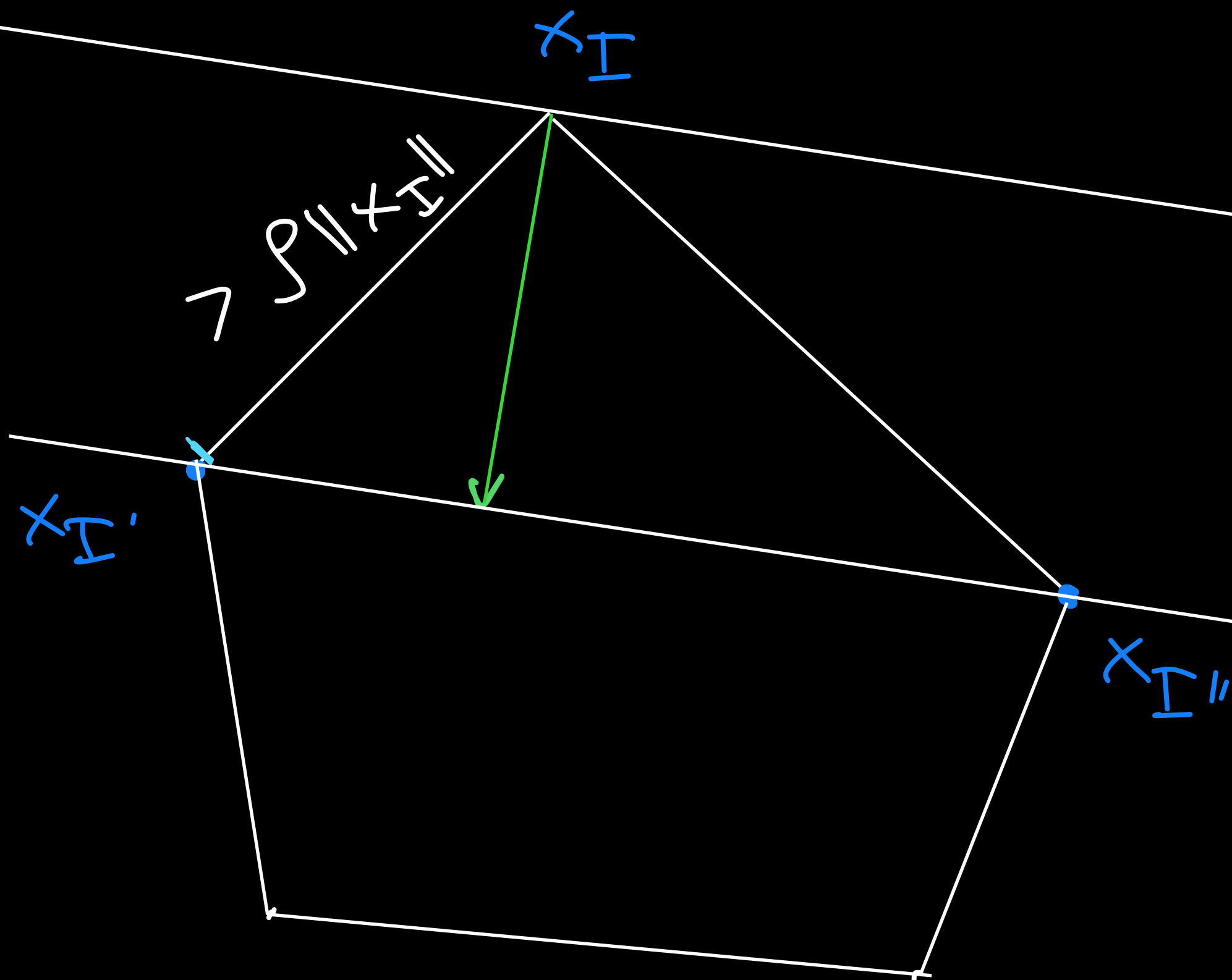
$\int_{[x_I, x_{I'}]} \|t\|^{-1} dt \geq \Omega(f)$



Case: 2) $\|x_I - x_{I'}\| > f \|x_I\|$

Full integral: $\int_{\partial Q} \|t\|^{-1} dt \leq O(d \log n)$

edge long $\Rightarrow$



$\int_{[x_I, x_{I'}]} \|t\|^{-1} dt \geq \Omega(f)$

$\Rightarrow \# \text{ case 2 edges } \leq O(d \log(n) f^{-1})$

edges of case 1 (large exterior angle)

$$\leq \frac{2\pi}{f\|x_I\|} \cdot L$$

edges of case 1 (large exterior angle)

$$\leq \frac{2\pi}{\mathcal{F} \|x_I\|} \cdot L$$

edges of case 2 (large perimeter integral consumption)

$$\leq O(d \log(n) \mathcal{F}^{-1})$$

edges of case 1 (large exterior angle)

$$\leq \frac{2\pi}{f \|x_I\|} \cdot \ell$$

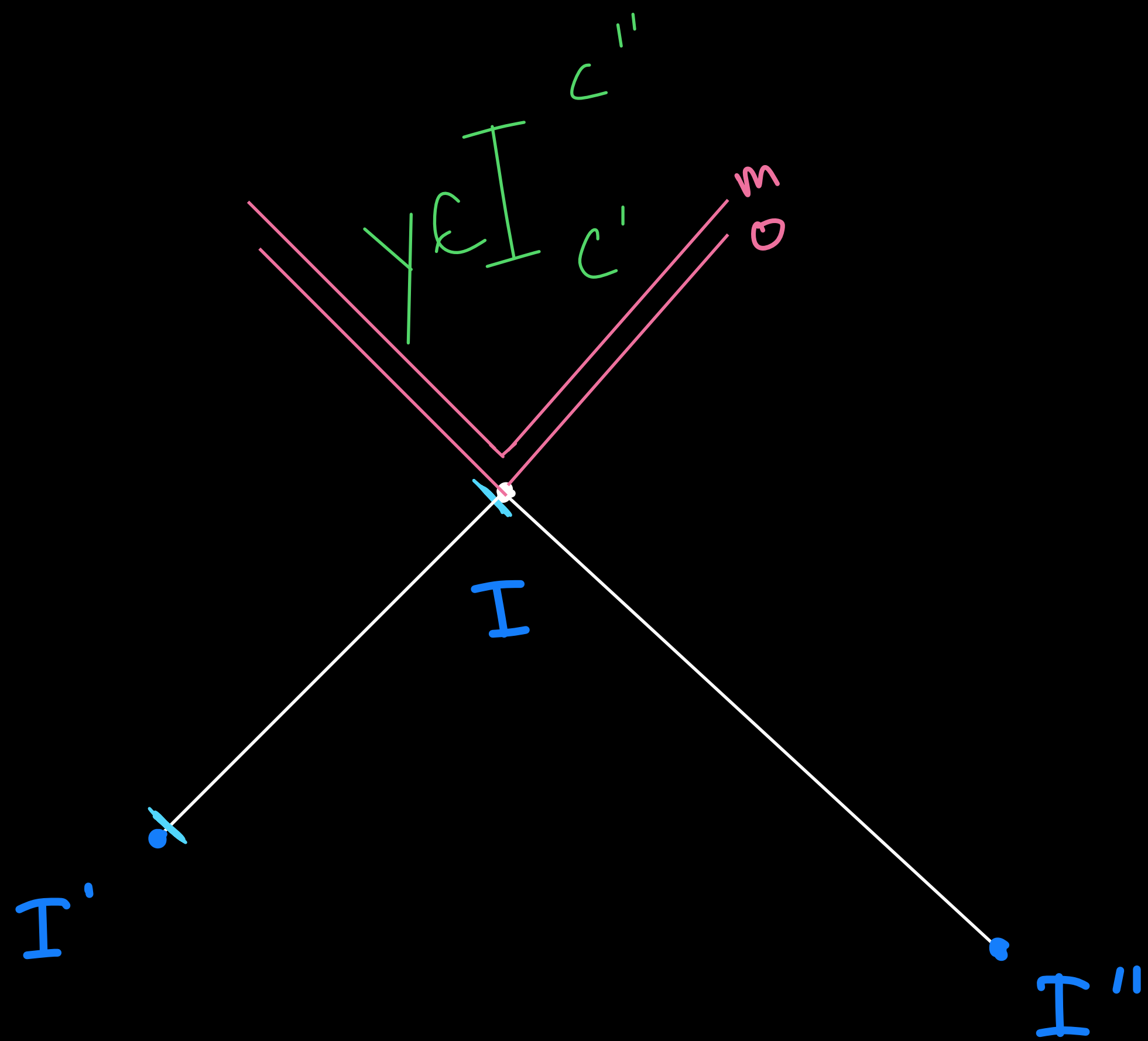
edges of case 2 (large perimeter integral consumption)

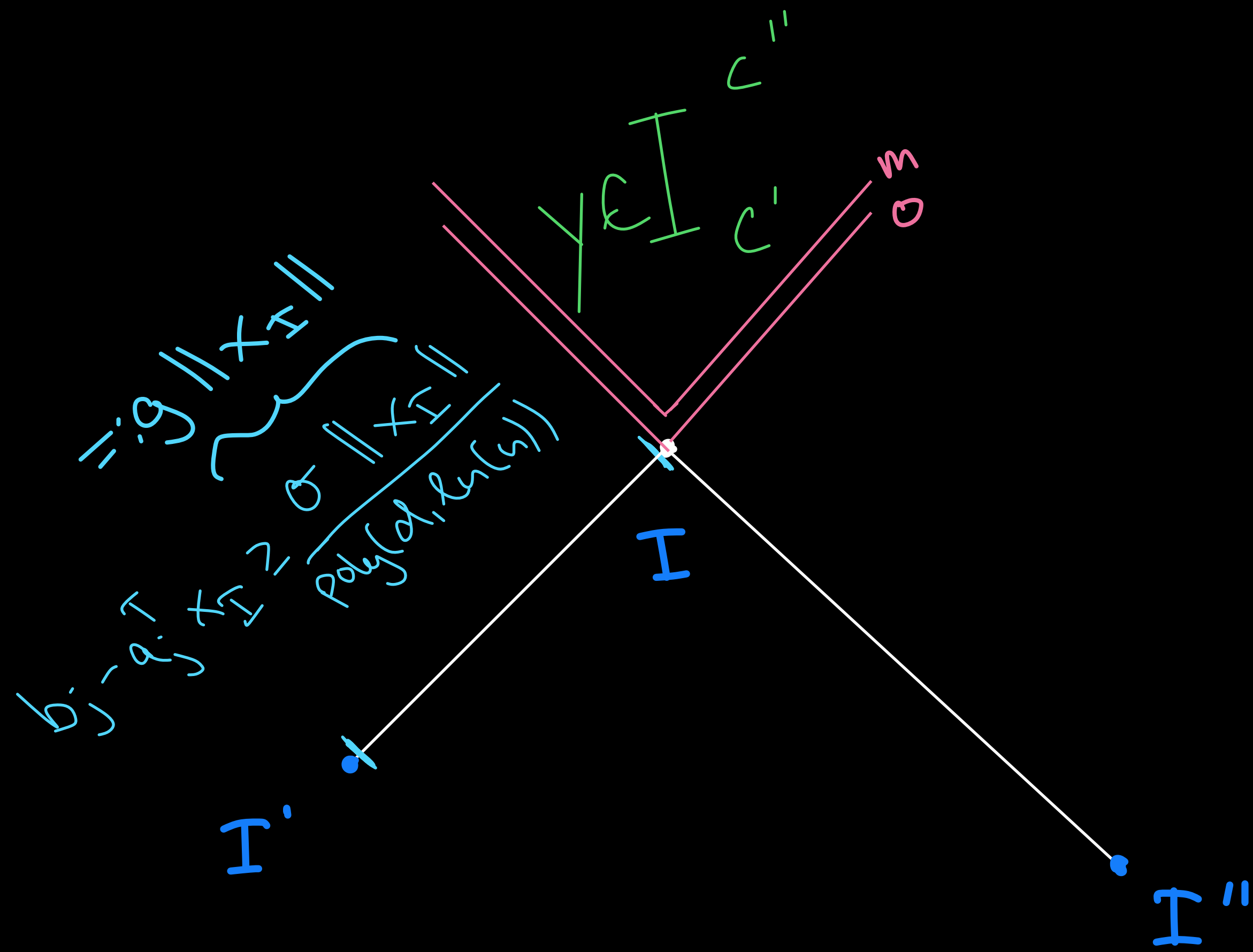
$$\leq O(d \log(n) f^{-1})$$

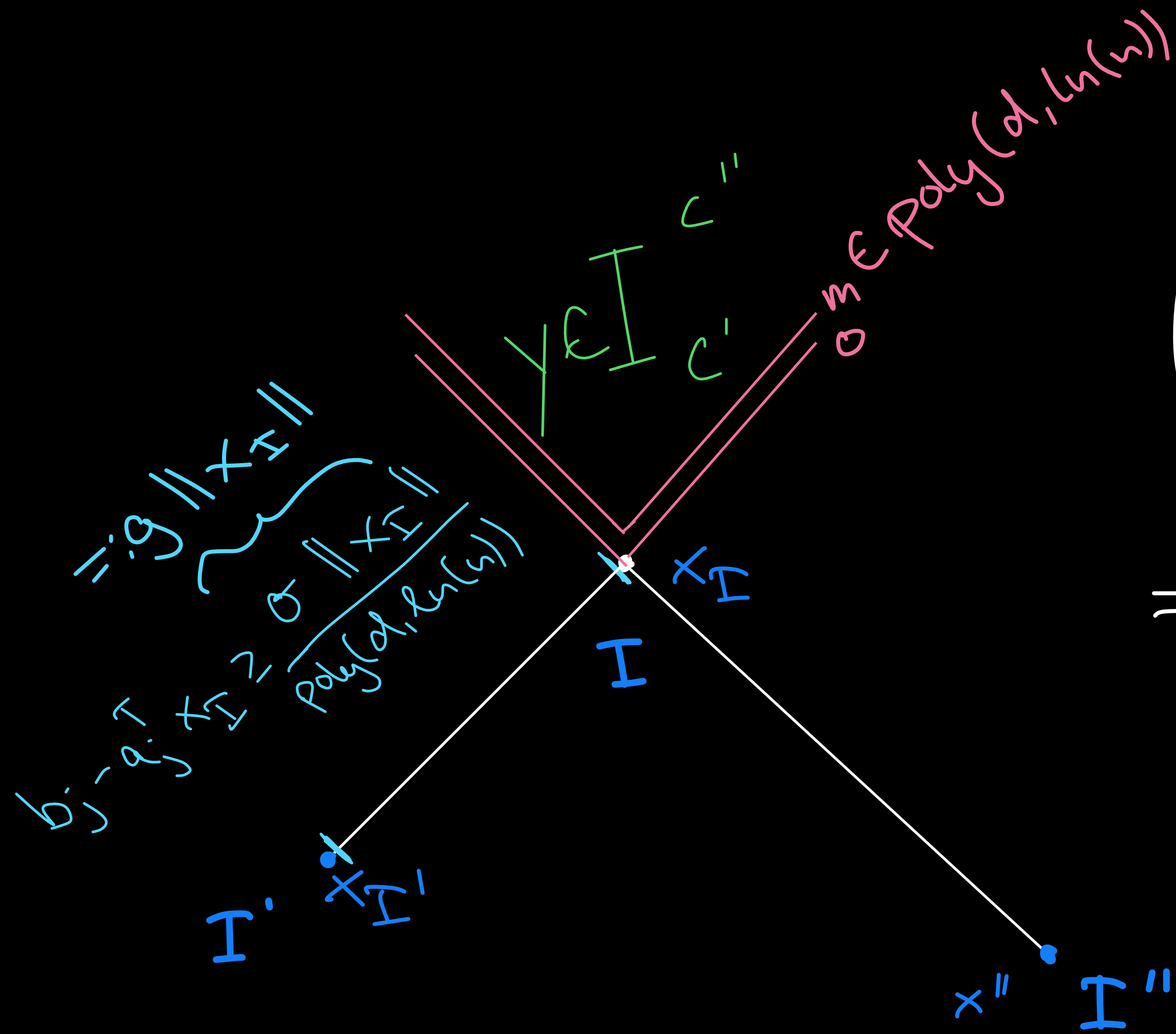
$$\Rightarrow \text{total \# edges} \leq O\left(\frac{\ell}{f \|x_I\|} + d \log(n) f^{-1}\right)$$

$$\text{For } f = \sqrt{\frac{\ell}{\|x_I\| d \log n}} \Rightarrow \leq O\left(\sqrt{\frac{\|x_I\| d \log n}{\ell}}\right)$$

How to determine L ?

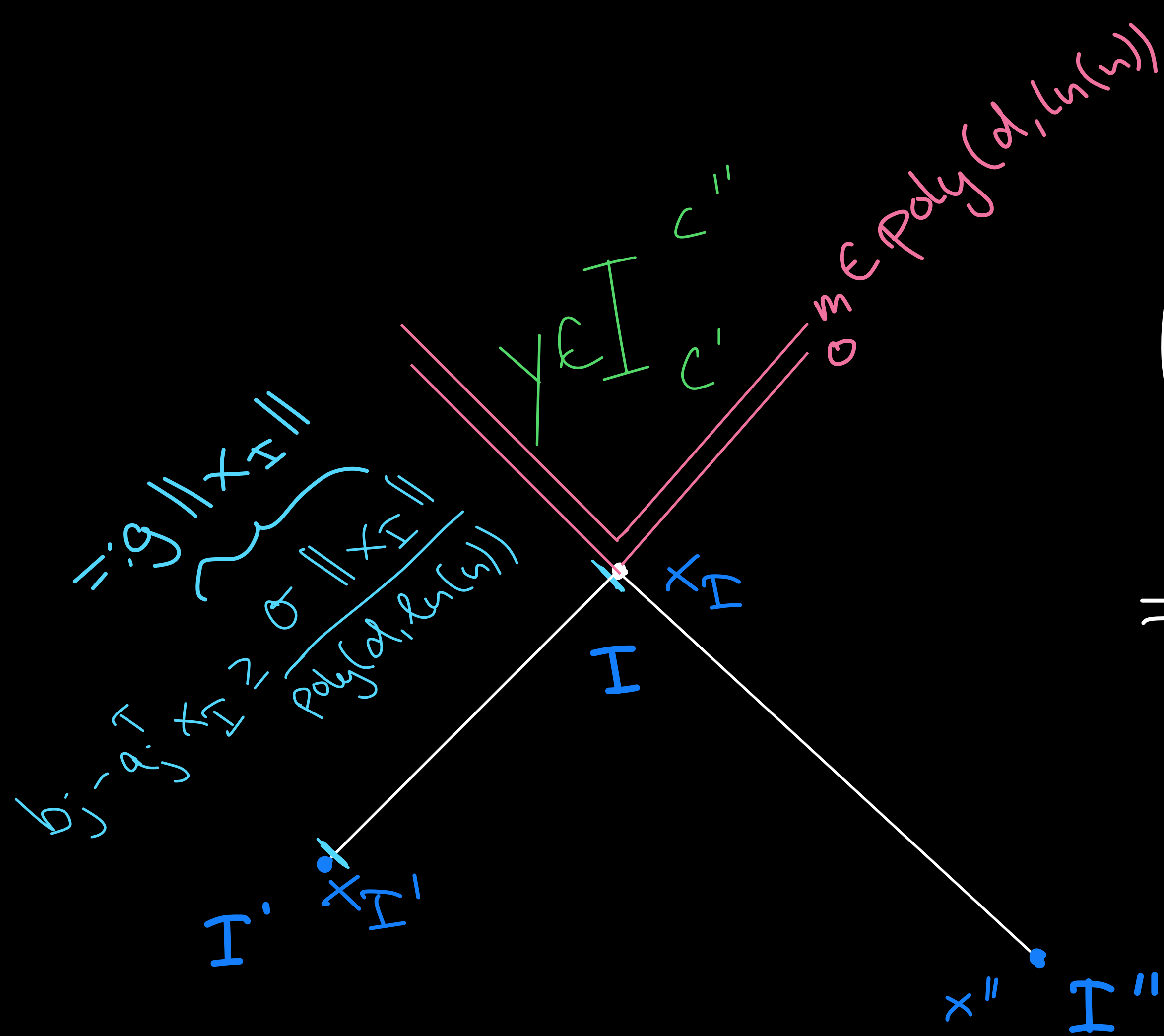






$$\left(\frac{y_I}{\|y_I\|} \right)^T (x_I - x_J)$$

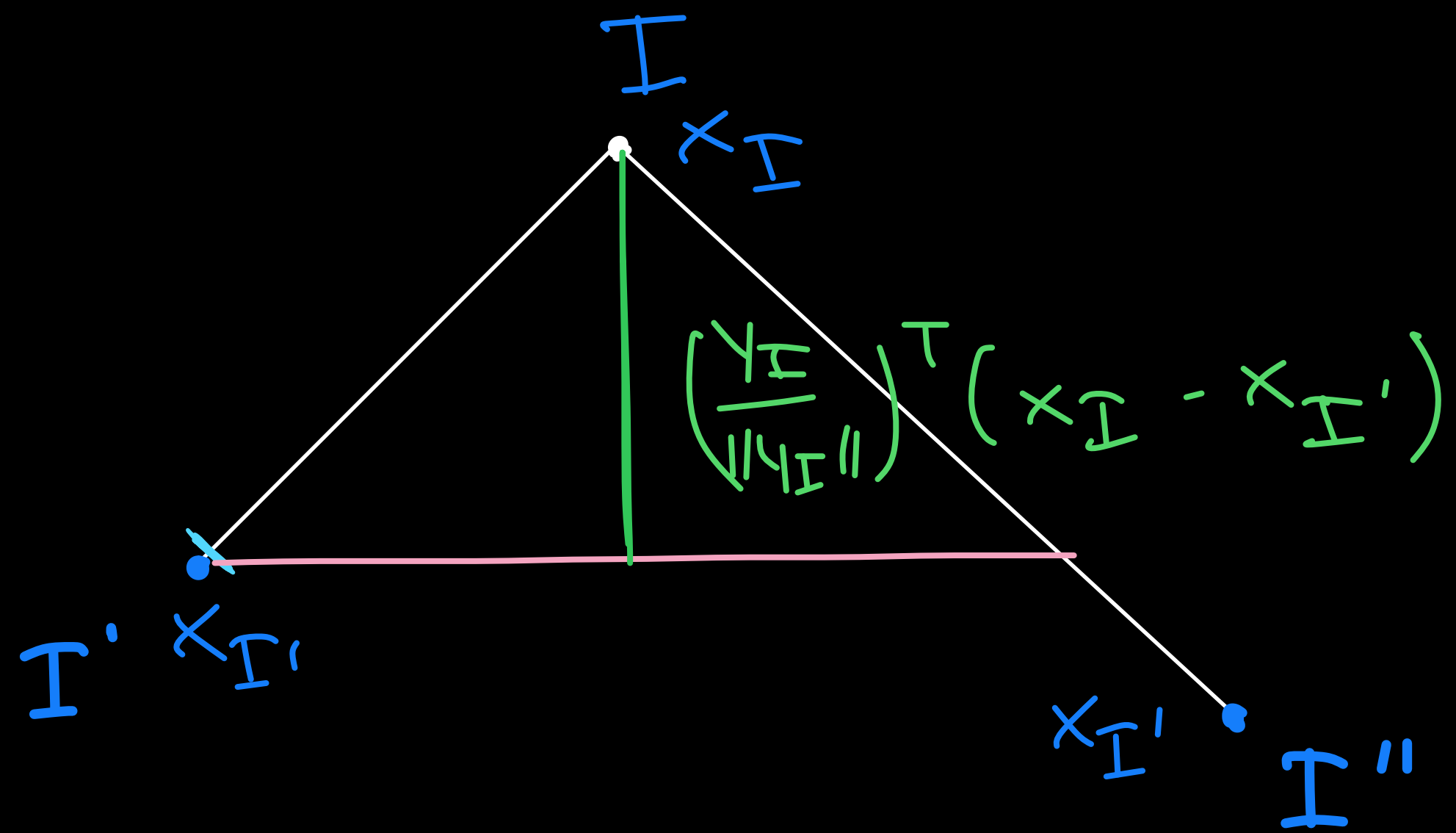
$$= \left(\frac{y_I}{\|y_I\|} \right)^T A_I^{-1} A_I (x_I - x_J)$$



$$\left(\frac{y_I}{\|y_I\|} \right)^T (x_I - x_J)$$

$$= \left(\frac{1}{\|y_I\|} \right)^T A_I^{-1} A_I (x_I - x_J)$$

$$\geq m \cdot \frac{\sigma \|x_I\|}{\text{poly}(d, \ln(n))} \cdot \frac{1}{\|y_I\|}$$



L

$$\geq \left(\frac{y_I}{\|y_I\|}\right)^T (x_I - x_{I'})$$

$$\geq m \cdot \frac{\sigma \|x_I\|}{\text{poly}(d, \ln(n))} \cdot \frac{1}{\|y_I\|}$$

$$\in O(\sigma \text{poly}(d, \ln(n) \|x_I\|))$$

$$\# \text{ edges of case 1 (large exterior angle)} \leq \frac{2\pi}{f \|x_I\|} \cdot L$$

$$\# \text{ edges of case 2 (large perimeter integral consumption)}$$

$$\leq O(d \log(n) f^{-1})$$

$$\Rightarrow \text{total \# edges} \leq O\left(\frac{L}{f \|x_I\|} + d \log(n) f^{-1}\right)$$

$$\text{For } f = \sqrt{\frac{L}{\|x_I\| d \log n}} \Rightarrow \leq O\left(\sqrt{\frac{\|x_I\| d \log n}{L}}\right)$$

$$\leq O\left(\sqrt{\frac{\|x_I\| d \log(n) \text{poly}(d, \ln(n))}{\|x_I\| \sigma}}\right) = O(\sigma^{-\frac{1}{2}} \text{poly}(d, \ln(n)))$$

Does smoothed analysis

explain efficiency of the

Simplex method?

Problem 1:

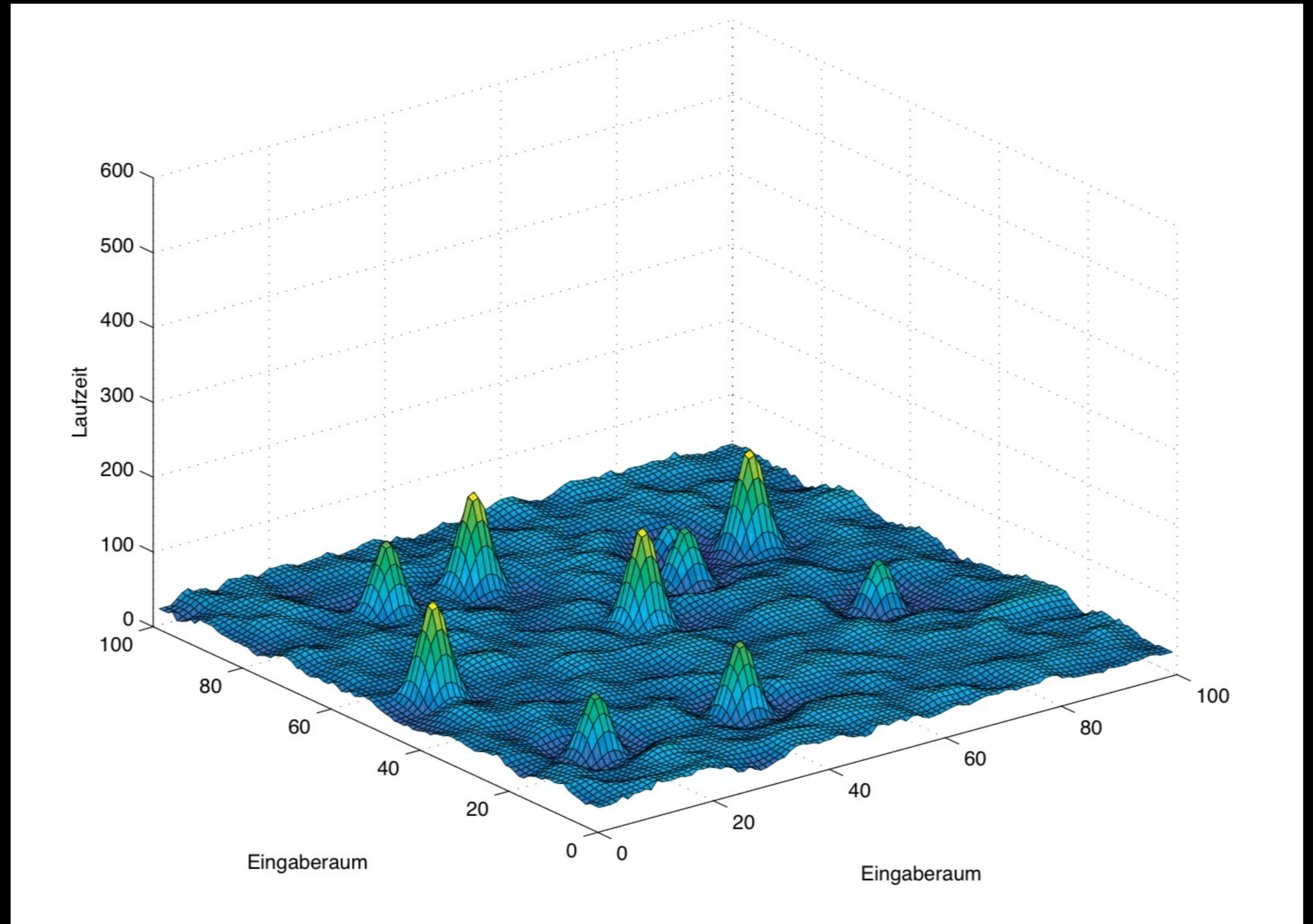
Real world LP:

$< 0.1\%$ of $A_{i,j} \neq 0$

Smoothed analysis

all $A_{i,j} \neq 0$

Problem 2:



K. H. Borgwardt: "Wie schnell arbeitet das Simplexverfahren normalerweise?"

Problem 3:

What are we modelling? What is σ ?

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- brittleness?

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What are we modelling? What is σ ?

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- brittleness?
- process within solver? $\rightarrow 10^{-6}$ solver tolerance?

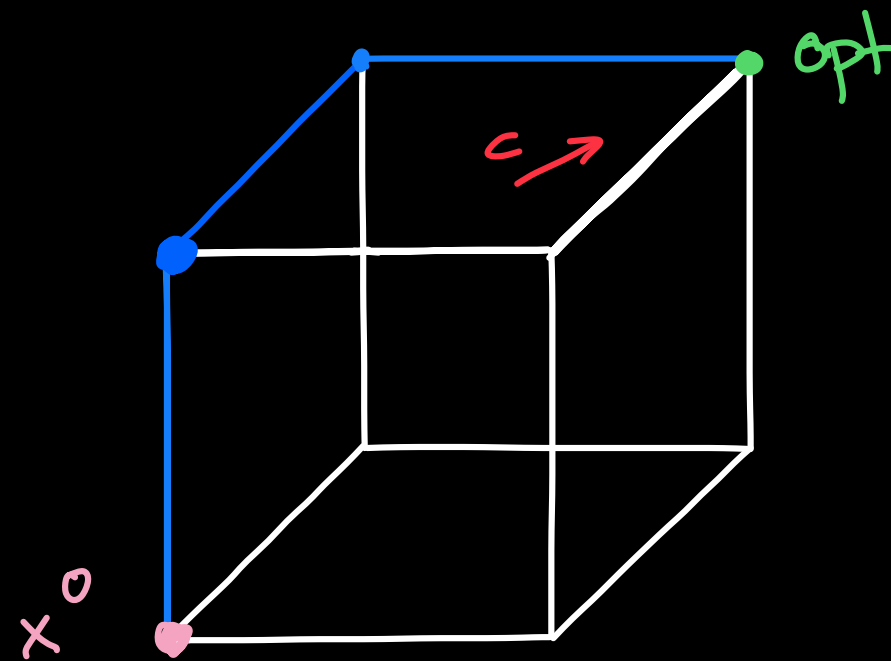
Problem 3:

What are we modelling? What is σ ?

- measurement error? $\rightarrow 10^{-2}$ sigfigs accuracy
- brittleness?
- process within solver? $\rightarrow 10^{-6}$ solver tolerance?
- floating point imprecision? 10^{-16} 64-bit IEEE 754?

Are we studying the simplex method?

- This geometry



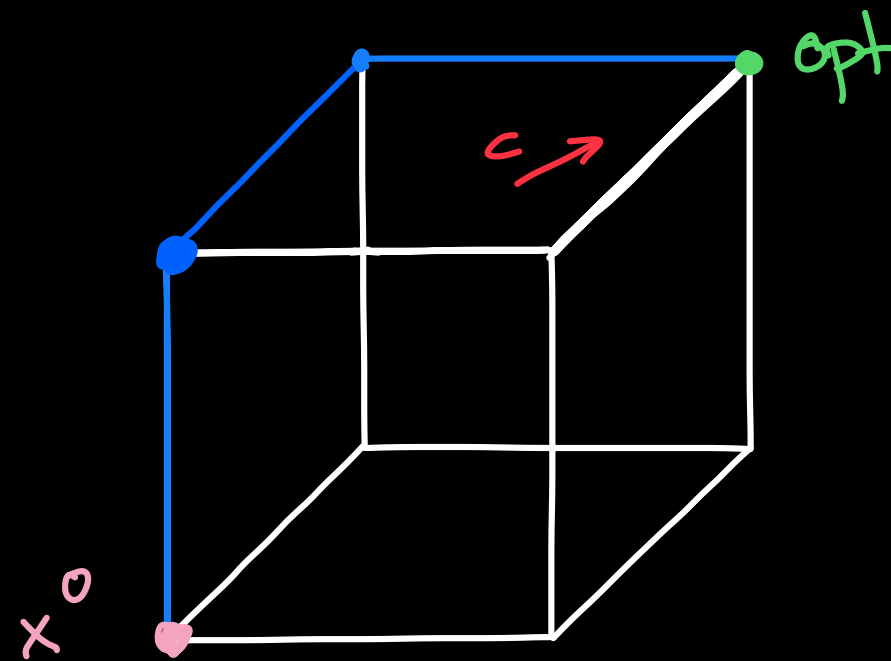
does not exist

- What exists :

{ numerical linear algebra

Are we studying the simplex method?

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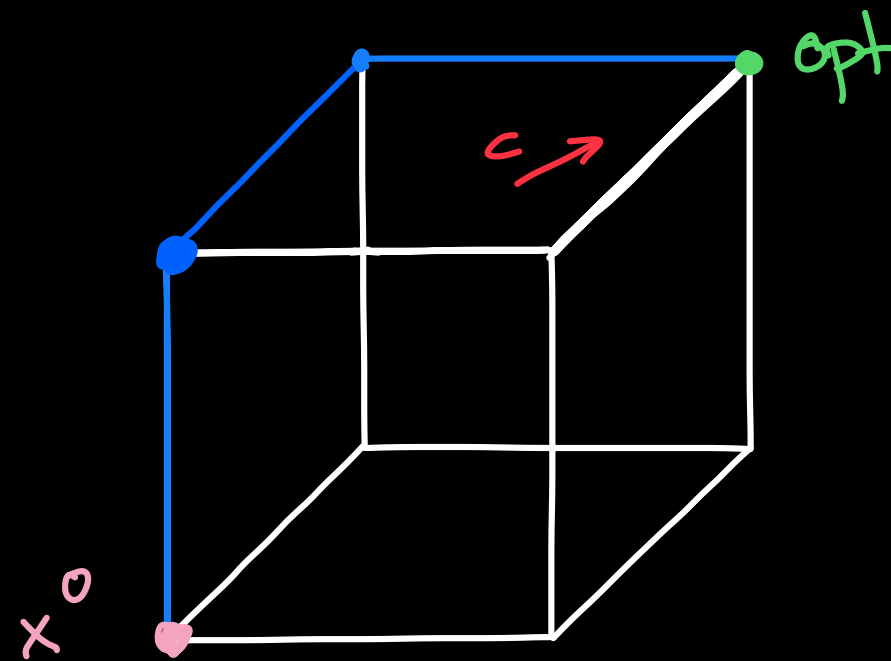
does not exist

- What exists :

{ numerical linear algebra
bound shifting

Are we studying the simplex method?

- This geometry



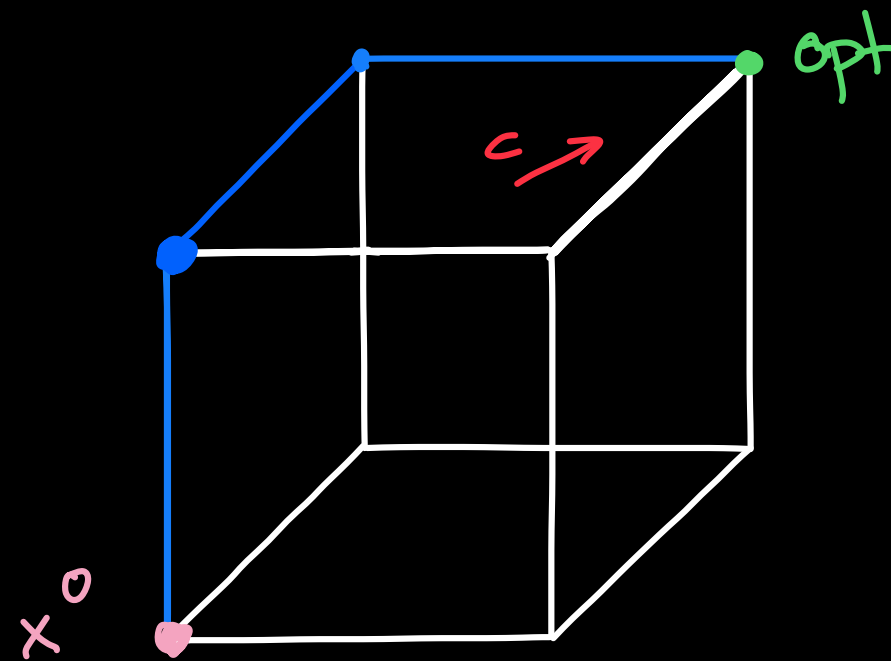
does not exist

- What exists :

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bound shifting
bound perturbations

Are we studying the simplex method?

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does not exist

- What exists :

{ numerical linear algebra
bound shifting
bound perturbations
(Harris) ratio test